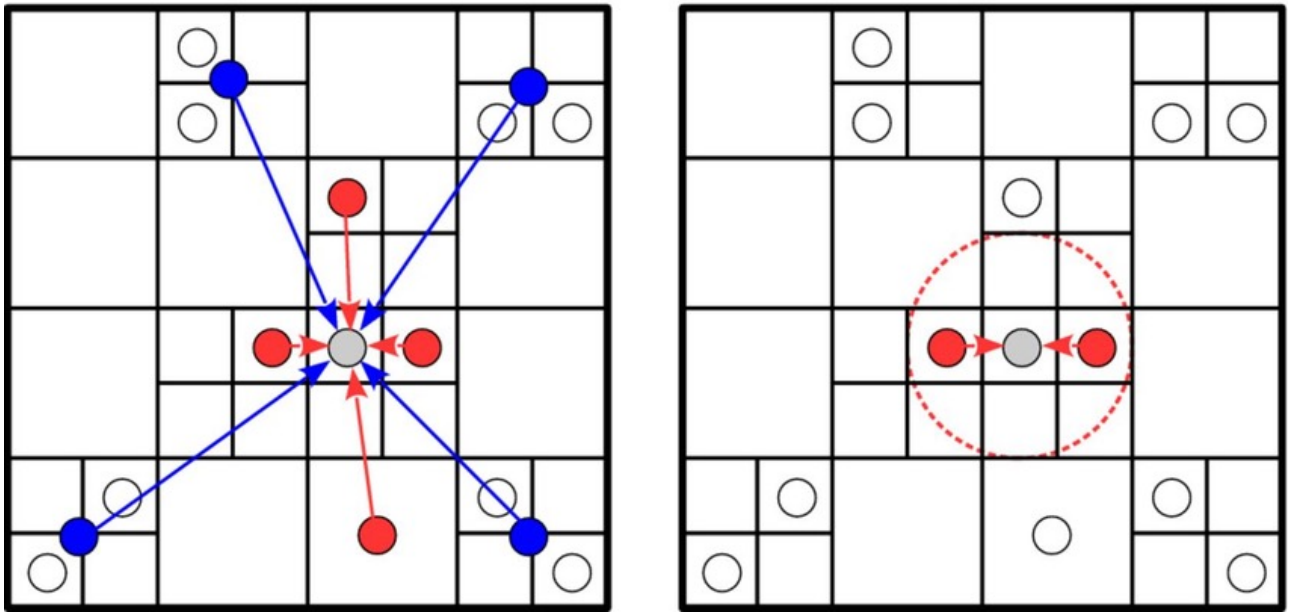
# *Force and Energy Computation*

*Week 1 : How MD works?*



2024 Winter Molecular Simulation Seminar

January 12th, 2024

Presenter: Seonghyeon Kang

Advisor: Prof. Chang Yun Son

# *Week 1 : How MD Works?*

## ***Basic Principle of Molecular Dynamics***

1. *Information of **Potential Energy** from **Force Field***
2. ***Calculation of Potential Energy***
3. *Computing Force, Displacement, Velocity, … from **Integration Algorithm***

## ***Two Kinds of Potential Energy(Brief Classification)***

1. ***Bonded Energy**(Bond, Angle, Torsion, Improper)*
   - *Easy and clear to compute(Low time complexity)*
   - *Short-range Interaction*
2. ***Nonbonded Energy**(Lennard-Jones, Electrostatic energy)*
   - *Difficult to compute(High time complexity)*
   - *Long-range interaction(Periodicity, Finite Size Effect)*

## *Contents*

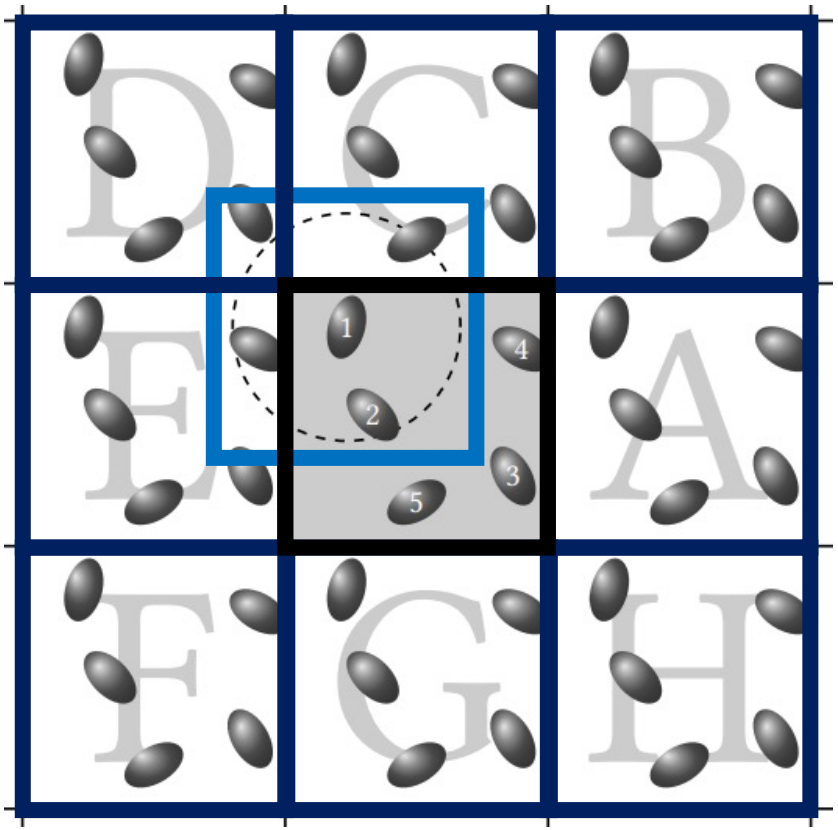

***1. Periodic Boundary Condition & Finite Size Effect***

***2. Computational Algorithms for Nonbonded Energy Calculation***

***3. Why we use PME(Particle-Mesh-Ewald)? & Artifacts of PME***

***4. GPU parallelization of Nonbonded Energy Computation***

# 1. Periodic Boundary Condition(PBC) & Finite Size Effect




*Allen, M. P., & Tildesley, D. J.*, Computer simulation of liquids. *Oxford university press.* **2017**

***Simulation Box***

***Periodic Image/Finite Size Convention***

*If we **don't apply periodic images**..*

*- **Surface effect** become dominant*

*- **PBC is applied to minimize it!***

*But PBC can make another artifacts*

*- It is called **Finite Size Effect***

*- Due to the **artificial external environments***

# 1. Periodic Boundary Condition(PBC) & Finite Size Effect

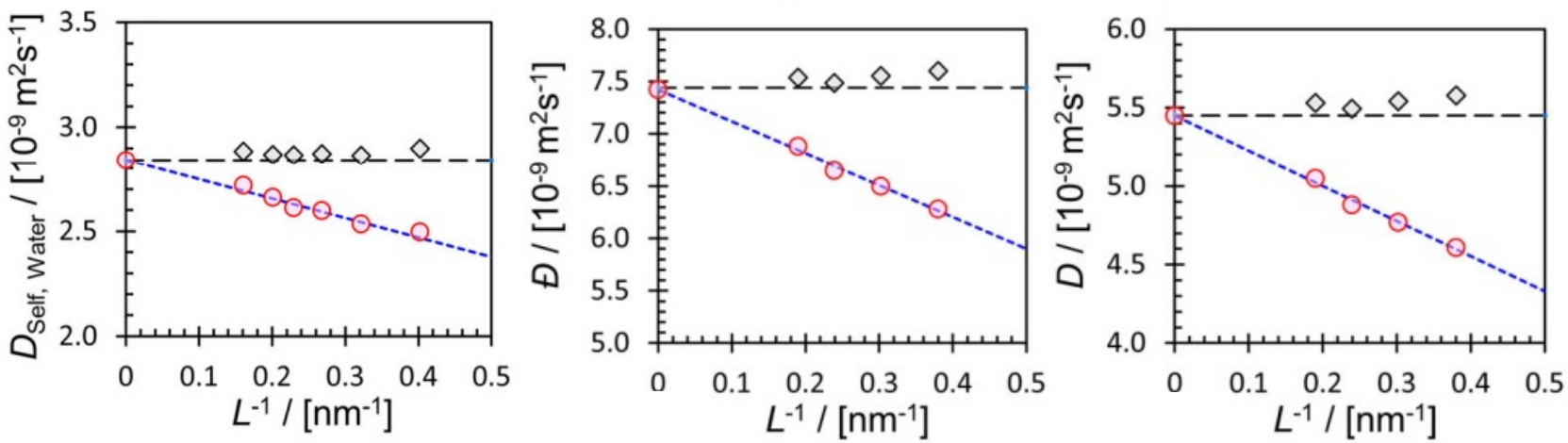
## Examples of Finite Size Effect

- *Diffusion Coefficient(And its correction)*

$$D_{\mathrm{Self}}^{\infty} = D_{\mathrm{Self}}^{\mathrm{MD}} + \frac{k_{\mathrm{B}} T \xi}{6\pi\eta L}$$ *- Self Diffusion Coefficient*

$$D_{\mathrm{Rot}}^{\infty} = D_{\mathrm{Rot}}^{\mathrm{MD}} + \frac{k_{\mathrm{B}} T}{6\eta V}$$ *- Rotational Diffusion Coefficient*

$$Đ^{\infty} = Đ^{\mathrm{MD}} + \left(\frac{1}{\Gamma}\right)\frac{k_{\mathrm{B}} T \xi}{6\pi\eta L} = Đ^{\mathrm{MD}} + \left(\frac{1}{\Gamma}\right) D^{\mathrm{YH}}$$ *- MS Diffusivity(Binary Mixture)*




*Celebi, A. T., Jamali, S. H., Bardow, A., Vlugt, T. J., & Moultos, O. A.,* Finite-size effects of diffusion coefficients computed from molecular dynamics: a review of what we have learned so far. *Mol. Sim.*, **2021**, *47*(10-11), 831-845.

## *1. Periodic Boundary Condition(PBC) & Finite Size Effect*

***Examples of Finite Size Effect***

- *Thermal Conductivity*
- *Solvation Energy*
- *…*

***The main reasons of these artifacts***

***1) Periodic images significantly affect the simulation box***

***2) Non-periodic nature of the real phenomenon***

***We have to set the proper size of simulation box to secure the accuracy and efficiency of computation!***

## 2. Computational Algorithms for Nonbonded Energy Calculation

$$U(R) = \sum_{bonds} k_r (r - r_{eq})^2 \quad \text{bond}$$

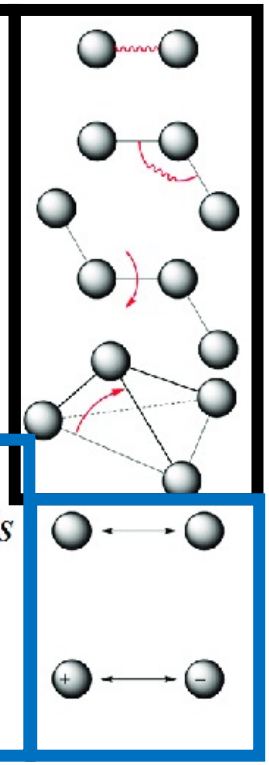
$$+ \sum_{angles} k_\theta (\theta - \theta_{eq})^2 \quad \text{angle}$$

$$+ \sum_{dihedrals} k_\phi (1 + \cos[n\phi - \gamma]) \quad \text{dihedral}$$

$$+ \sum_{impropers} k_\omega (\omega - \omega_{eq})^2 \quad \text{improper}$$

$$+ \sum_{i<j}^{atoms} \varepsilon_{ij} \left[ \left( \frac{r_m}{r_{ij}} \right)^{12} - 2 \left( \frac{r_m}{r_{ij}} \right)^{6} \right] \quad \text{van der Waals}$$

$$+ \sum_{i<j}^{atoms} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \quad \text{electrostatic}$$



**Bonded Energy**

- *Easy and clear to compute*
- *Short-range Interaction*
- *Low time complexity*

**Nonbonded Energy**

- *Difficult to compute*
- *Long-range Interaction*
- *High time complexity*

***Van-der Walls energy(Rapidly decay, ~ $r^{-6}$)***

*- Cutoff energy*

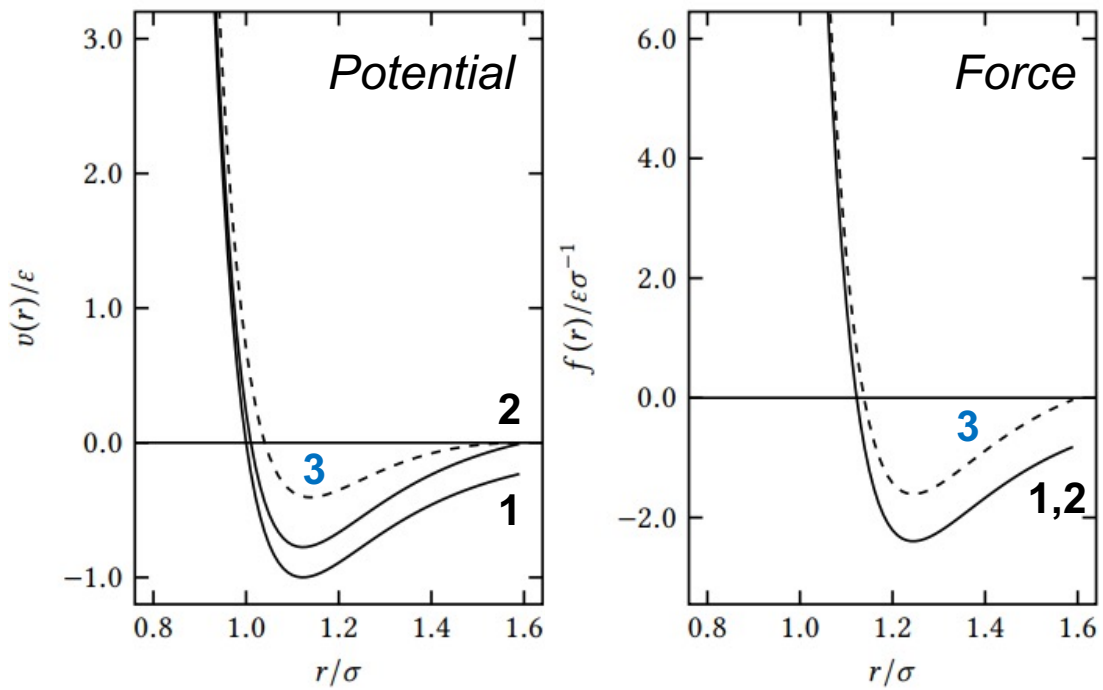
***Electrostatic energy(Slowly decay, ~ $r^{-1}$)***

*- Cutoff unavailable*

*- PME, PPPM, Reaction Field, FMM, …*

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Van-der Walls energy




1. Lennard-Jones Potential

$$v^{\mathrm{LJ}}(r) = 4\epsilon\left[(\sigma/r)^{12} - (\sigma/r)^{6}\right]$$

2. Shifted Lennard-Jones

$$v^{\mathrm{S}}(r_{ij}) = \begin{cases} v(r_{ij}) - v_{\mathrm{c}} & r_{ij} \le r_{\mathrm{c}} \\ 0 & r_{ij} > r_{\mathrm{c}} \end{cases}$$

### 3. Shift-force potential

$$v^{SF}(r_{ij}) = \begin{cases} v(r_{ij}) - v_c - (r_{ij} - r_c)\left(\dfrac{dv(r_{ij})}{dr_{ij}}\right) & (r_{ij} \le r_c) \\ 0 \quad (r_{ij} \ge r_c) \end{cases}$$

1. Zero potential after $r_c$
2. Continuous & Differentiable potential function

*Allen, M. P., & Tildesley, D. J.*, Computer simulation of liquids. *Oxford university press.* **2017**

## 2. Computational Algorithms for Nonbonded Energy Calculation

### Electrostatic Interaction

#### Direct Coulomb Summation

$$E_{elec} = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}^{N}\frac{q_i q_j}{r_{ij}}$$

***Exact summation*** *of all coulombic interaction*
*The most accurate, but also* ***the most expensive one(O(N²))***
*Moreover, it has* ***low scalability*** *to parallelize this algorithm*

#### Alternative approach to make efficient algorithms

***Particle Mesh Ewald(PME) Method***

***Particle – Particle/Particle-Mesh(PPPM) Method***

***Reaction Field Method***

***Fast Multipole Method***

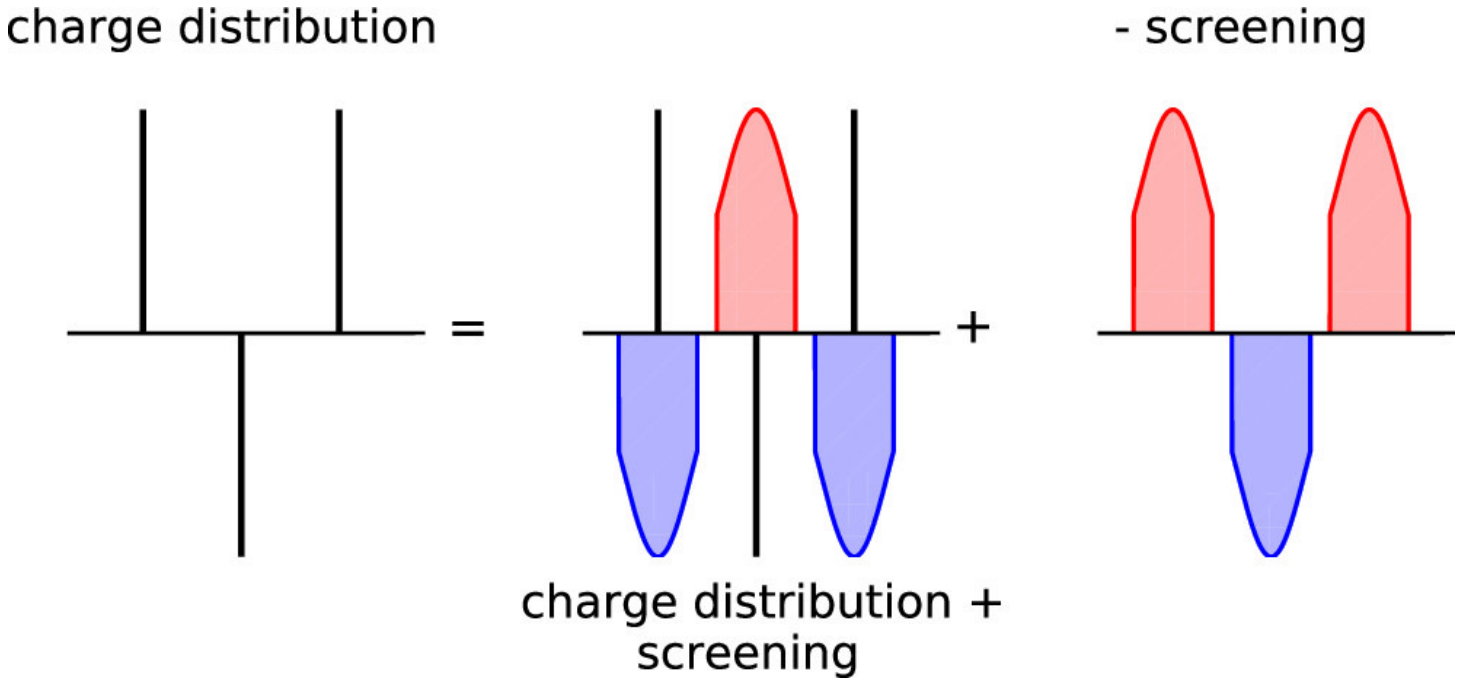
*(OpenMM utilizes sPME & Reaction Field Method)*

## 2. Computational Algorithms for Nonbonded Energy Calculation

### Electrostatic Interaction

#### Particle Mesh Ewald(PME) Method(Idea)

**Sum of Coulombic energy = real space sum + reciprocal space sum**




*Stenberg, S., & Stenqvist, B.*, An exact Ewald summation method in theory and practice. *The Journal of Physical Chemistry A,* **2020***,124(*19*),* 3943-3946.

*Narrow distribution in real space : Broad distribution of reciprocal space*
*Broad distribution in real space : Narrow distribution of reciprocal space*

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Electrostatic Interaction

### Particle Mesh Ewald(PME) Method(Formalism & Performance)

$$E_{\text{dir}} = \frac{1}{2}\sum_{n}^{*}\sum_{i,j=1}^{N}\frac{q_j q_j \operatorname{erfc}(\beta|\mathbf{r}_j - \mathbf{r}_i + \mathbf{n}|)}{|\mathbf{r}_j - \mathbf{r}_i + \mathbf{n}|},$$

*Direct space summation*

$$E_{\text{rec}} = \frac{1}{2\pi V}\sum_{\mathbf{m}\neq 0}\frac{\exp(-\pi^2\mathbf{m}^2/\beta^2)}{\mathbf{m}^2}S(\mathbf{m})S(-\mathbf{m}),$$

*Reciprocal space summation*

$$E_{\text{corr}} = -\frac{1}{2}\sum_{(i,j)\in M}\frac{q_i q_j \operatorname{erf}(\beta|\mathbf{r}_i - \mathbf{r}_j|)}{|\mathbf{r}_i - \mathbf{r}_j|} - \frac{\beta}{\sqrt{\pi}}\sum_{i=1}^{N} q_i^2$$

*Error correction(Self-interaction)summation*

$$S(\mathbf{m}) = \sum_{j=1}^{N} q_j \exp(2\pi i \mathbf{m}\cdot\mathbf{r}_j) = \sum_{j=1}^{N} q_j \exp[2\pi i(m_1 s_{1j} + m_2 s_{2j} + m_3 s_{3j})]$$ *(Structure Factor)*

*Essmann, U., Perera, L., Berkowitz, M. L., Darden, T., Lee, H., & Pedersen, L. G.,* A smooth particle mesh Ewald method. *The Journal of chemical physics*, **1995**, *103*(19), 8577-8593.

***Time Complexity : O(NlogN)***
***Accuracy : Medium accuracy, Depending on $\alpha$(Ewald splitting parameter)***
***Scalability : High(3D Discrete Fast Fourier Transform)***

*George, A., Mondal, S., Purnaprajna, M., & Athri, P.,* Review of Electrostatic Force Calculation Methods and Their Acceleration in Molecular Dynamics Packages Using Graphics Processors. *ACS omega, **2022**, 7(37),* 32877-32896.

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Electrostatic Interaction

### Particle Mesh Ewald(PME) Method(PME vs sPME)

$$S(\mathbf{m}) = \sum_{j=1}^{N} q_j \exp(2\pi i \mathbf{m}\cdot\mathbf{r}_j) = \sum_{j=1}^{N} q_j \exp[2\pi i(m_1 s_{1j} + m_2 s_{2j} + m_3 s_{3j})]$$ *(Structure Factor)*

$$\exp(2\pi i \mathbf{m}\cdot\mathbf{r}) = \exp\left(2\pi i \frac{m_1 u_1}{K_1}\right)\cdot\exp\left(2\pi i \frac{m_2 u_2}{K_2}\right)\cdot\exp\left(2\pi i \frac{m_3 u_3}{K_3}\right)$$

### Approximation of Structure Factor I : Lagrangian Interpolation(PME)

$$\exp\left(2\pi i \frac{m_\alpha}{K_\alpha} u_\alpha\right) \approx \sum_{k=-\infty}^{\infty} W_2(u_\alpha - k)\cdot\exp\left(2\pi i \frac{m_\alpha}{K_\alpha} k\right)$$

$$W_{2p}(u) = \frac{\Pi_{j=-p, j\neq k}^{p-1}(u+j-k)}{\Pi_{j=-p, j\neq k}^{p-1}(j-k)}$$

for $k \leqslant u \leqslant k+1, k = -p, -p+1, \ldots, p-1$

### Approximation of Structure Factor II : Cardinal B-Spline(sPME)

$$\exp\left(2\pi i \frac{m_i}{K_i} u_i\right) \approx b_i(m_i) \sum_{k=-\infty}^{\infty} M_n(u_i - k)\cdot\exp\left(2\pi i \frac{m_i}{K_i} k\right)$$

**Implemented in OpenMM**

$$M_n(u) = \frac{u}{n-1} M_{n-1}(u) + \frac{n-u}{n-1} M_{n-1}(u-1)$$

$$b_i(m_i) = \exp(2\pi i(n-1)m_i/K_i) \times \left[\sum_{k=0}^{n-2} M_n(k+1)\exp(2\pi i m_i k/K_i)\right]^{-1}$$

*Essmann, U., Perera, L., Berkowitz, M. L., Darden, T., Lee, H., & Pedersen, L. G.,* A smooth particle mesh Ewald method. *The Journal of chemical physics*, **1995**, *103*(19), 8577-8593.

## 2. Computational Algorithms for Nonbonded Energy Calculation

### Electrostatic Interaction

#### Particle – Particle/Particle – Mesh(PPPM) Method

*Based on Ewald sum(dividing short range/long range interaction + FFT)*

***Short range : particle-particle & Long range : particle-mesh***

$$\psi_s(\zeta_{ij}) = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{r_{ij}} - \frac{1}{70a}\sum_{n=-1}^{7} C_n \zeta_{ij}^n\right) \zeta_{ij} < 2$$

*Short range interaction*

$$\hat{\psi}(k) = \hat{\gamma}\frac{(k)^2}{\epsilon_0 k^2}\hat{\rho}(k) = \hat{G}(k)\hat{\rho}(k)$$

*Long range interaction*

$$\zeta_{ij} = \frac{2r_{ij}}{a} \quad \gamma(r) = \begin{cases} \frac{48}{\pi a^4}\left(\frac{a}{2} - r\right) & r < \frac{a}{2} \\ 0 & r > \frac{a}{2} \end{cases}$$

***Time Complexity : O(NlogN)***

***Accuracy : High accuracy, More than PME***

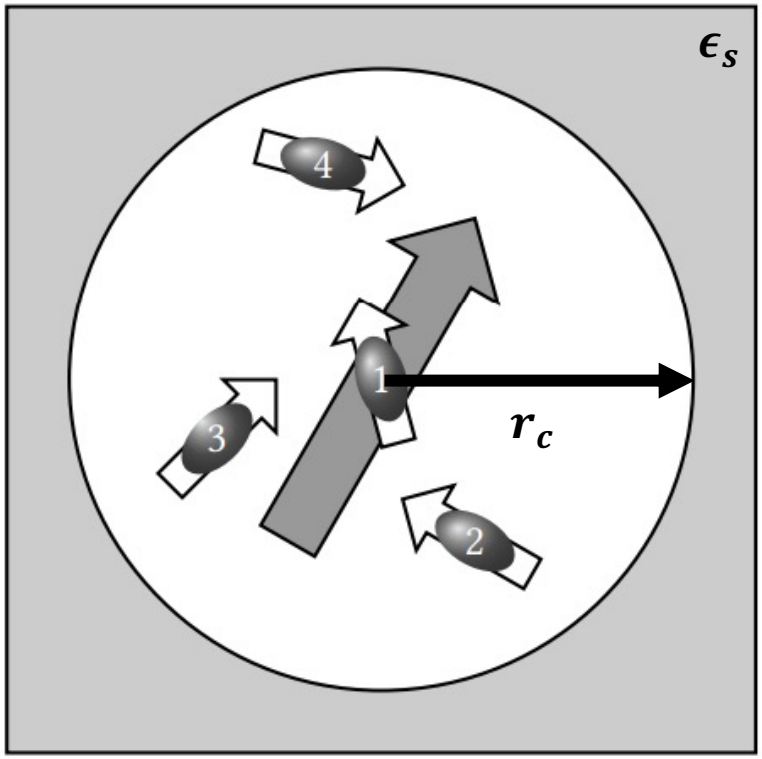
***Scalability : Medium(Increase of long-range part make lower scalability)***

*George, A., Mondal, S., Purnaprajna, M., & Athri, P.,* Review of Electrostatic Force Calculation Methods and Their Acceleration in Molecular Dynamics Packages Using Graphics Processors. *ACS omega,* ***2022****, 7(37),* 32877-32896.

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Electrostatic Interaction

### Reaction Field Method(Idea)




**Spherical Truncation($r < r_c$)**
**+ Dielectric Continuum Outside**

*Explicit interaction weighting*

$$f(r_{ij}) = \begin{cases} 1.0 & r_{ij} < r_t \\ (r_c - r_{ij})/(r_c - r_t) & r_t \le r_{ij} \le r_c \\ 0.0 & r_c < r_{ij} \end{cases}$$

$(r_t \sim 0.95 r_c)$

*Reaction field acting on molecule i*

$$\mathcal{E}_i = \frac{2(\epsilon_s - 1)}{2\epsilon_s + 1} \frac{1}{r_c^3} \sum_{j \in \mathcal{R}} \boldsymbol{\mu}_j$$

*Allen, M. P., & Tildesley, D. J.*, Computer simulation of liquids. *Oxford university press.* **2017**

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Electrostatic Interaction

### Reaction Field Method(Formalism)

$$\mathcal{V}^{qq} = \frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j,\, I\neq J\\ R_{IJ}<r_c}}^{N} q_i q_j \left[\frac{1}{r_{ij}} + \left(\frac{\epsilon_s - 1}{2\epsilon_s + 1}\frac{r_{ij}^2}{r_c^3} + C\right)\right] + \mathcal{V}_{\text{Born}} + \mathcal{V}_{\text{self}}$$

$$\mathcal{V}_{\text{Born}} = -\frac{1}{2}\left(\frac{\epsilon_s - 1}{\epsilon_s}\right)\frac{1}{r_c}\sum_{i=1}^{N} q_i \sum_{\substack{j\\ R_{IJ}<r_c}}^{N} q_j \qquad \mathcal{V}_{\text{self}} = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\in I}^{N} q_i q_j \left(\frac{\epsilon_s - 1}{2\epsilon_s + 1}\frac{r_{ij}^2}{r_c^3} + C\right) \qquad C = -\frac{1}{r_c}\left[1 + \frac{\epsilon_s - 1}{2\epsilon_s + 1} - \frac{\epsilon_s - 1}{\epsilon_s}\right]$$

*Linearized Poisson-Boltzmann Equation*

$$\mathcal{V}^{qq} = \frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j\neq i\\ r_{ij}<r_c}} q_i q_j \left[\frac{1}{r_{ij}} - \frac{(1+B_1)r_{ij}^2}{2r_c^3}\right]$$

$$B_1 = \frac{(1-4\epsilon_s)(1+\kappa r_c) - 2\epsilon_s(\kappa r_c)^2}{(1+2\epsilon_s)(1+\kappa r_c) + \epsilon_s(\kappa r_c)^2}$$

***Time Complexity : O(N)***
***Accuracy : Low accuracy***
***Scalability : High***

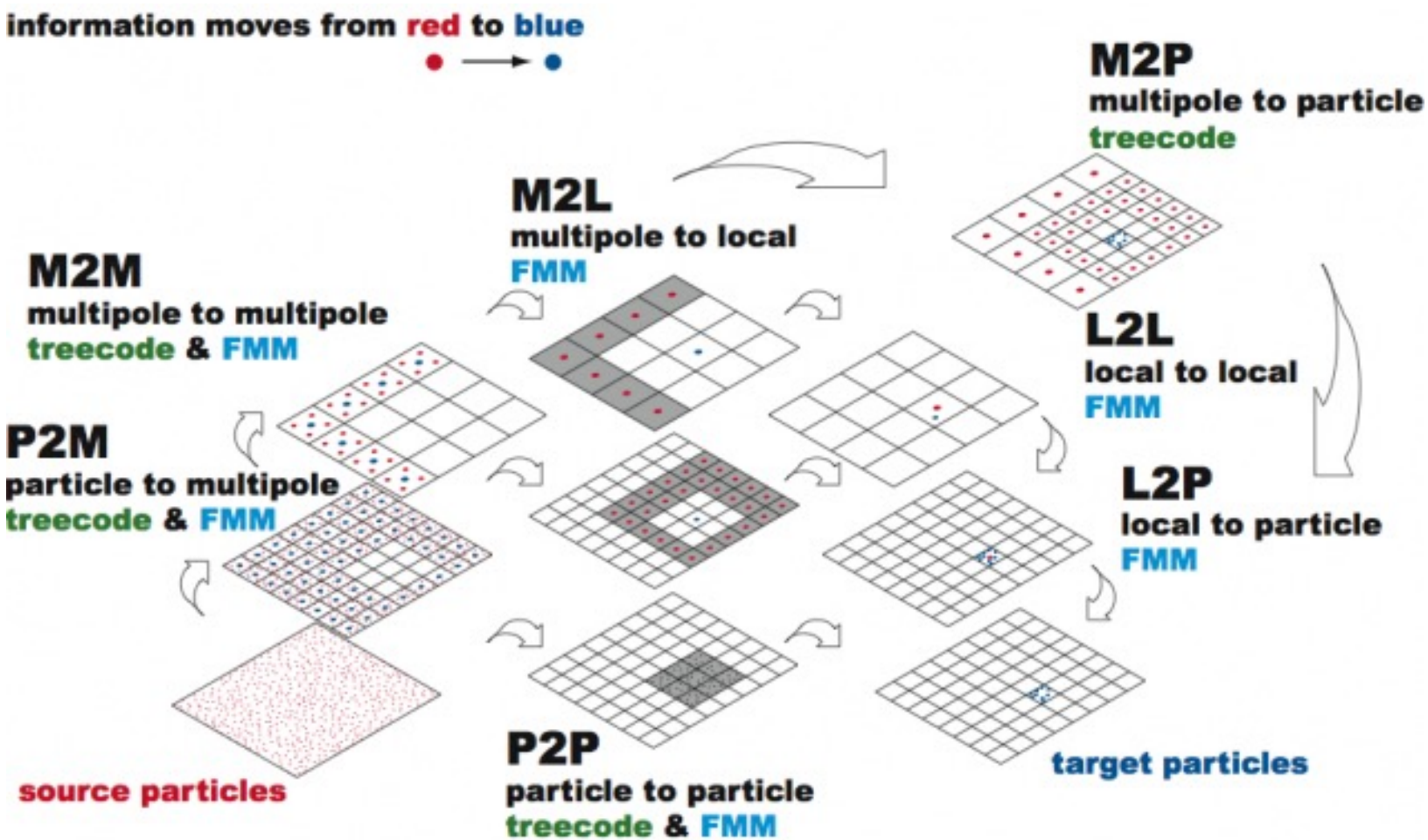
***Reaction Field Method is Implemented in OpenMM***

*Allen, M. P., & Tildesley, D. J.*, Computer simulation of liquids. *Oxford university press.* **2017**

# 2. Computational Algorithms for Nonbonded Energy Calculation

## Electrostatic Interaction

### Fast Multipole Method(Idea)




*Yokota, R., & Barba, L. A.,* GPU Computing Gems Emerald Edition, chapter 9: Treecode and fast multipole method for N-body simulation with CUDA, **2011**.

## 2. Computational Algorithms for Nonbonded Energy Calculation

### Electrostatic Interaction

#### Fast Multipole Method(Formalism & Performance)

*Potential*

$$\phi(\mathbf{r}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{m=\ell} M_{\ell,m} G_{\ell,m}(\mathbf{r}) \qquad G_{\ell,m}(\mathbf{r}) = \frac{(-1)^{\ell-m}(\ell-m)!}{r^{\ell+1}} \exp(im\varphi) P_{\ell,m}(\cos\theta).$$

*Solution*

$$M_{\ell,m} = \sum_{i=1}^{N} q_i F^*_{\ell,m}(-\mathbf{r}_i), \quad F_{\ell,m}(\mathbf{r}) = \frac{(-1)^{\ell-m}\, r^{\ell}}{(\ell+m)!} \exp(im\varphi) P_{\ell,m}(\cos\theta).$$

***Time Complexity : O(N), top 10 algorithms in 20th century***
***Accuracy : Medium accuracy***
***Scalability : High, One of the most scalable algorithm***

***FMM is Implemented in GROMACS***

## *3. Why we use PME(Particle-Mesh-Ewald)? & Artifacts of PME*

### *Why we use PME nowadays?*

*1. Low complexity(O(NlogN))*

*2. Quite high accuracy*

*3. High scalability*

*4. Traditionally utilized*

*5. Easy to parametrize*

### *Limits & Artifacts of PME*

*1. Only appliable in charge neutral system*

*2. In non-neutral system, uniform background charge distribution to compensate the charge of the system is applied.*

*3. The background charge induces artifacts related to dielectric constant in inhomogeneous system*

*Ex) Over-stabilization of ions in low-dielectric medium*

## 3. Why we use PME(Particle-Mesh-Ewald)? & Artifacts of PME

### Limits & Artifacts of PME

1. Only appliable in charge neutral system

2. In non-neutral system, uniform background charge distribution to compensate the charge of the system is applied.

$$E_{elec} = E_{real} + E_{recip} + E_{back} - E_{self}$$

$$E_{real} = \frac{1}{2}\sum\sum_{i \neq j} q_i \phi_{real}$$

$$E_{back} = \frac{1}{2}\int_V \rho_{back}\phi_{real}\, d\boldsymbol{r} = \frac{\pi q_{tot}^2}{2V\alpha}$$

$$E_{recip} = \frac{1}{2}\sum_k \tilde{\rho}_{recip}\tilde{\phi}_{recip}\, exp(i\boldsymbol{kr_i})$$

$$= \frac{1}{2}\sum_k (\tilde{\rho}_{back} + \tilde{\rho}_{smooth})\tilde{\phi}_{recip}\, exp(i\boldsymbol{kr_i})$$

$$\boxed{= \frac{1}{2}\sum_k \tilde{\rho}_{smooth}\tilde{\phi}_{recip}\, exp(i\boldsymbol{kr_i})}$$

Infinite sum
Low convergence

$\rho_{back}$ is constant, $\tilde{\rho}_{back}\tilde{\phi}_{recip} = 0\ for\ k \neq 0$

Also, $\tilde{\rho}_{recip}\tilde{\phi}_{recip}(k=0) = \int \rho_{recip}\, dV = 0$

**Ewald sum will converge, but the energy will be depend on non-physical parameter $\alpha$**

## *3. Why we use PME(Particle-Mesh-Ewald)? & Artifacts of PME*

### *Limits & Artifacts of PME*

*3. The background charge induces artifacts related to dielectric constant in inhomogeneous system*

*Utilizing PME method with non-neutral system : Technically possible*

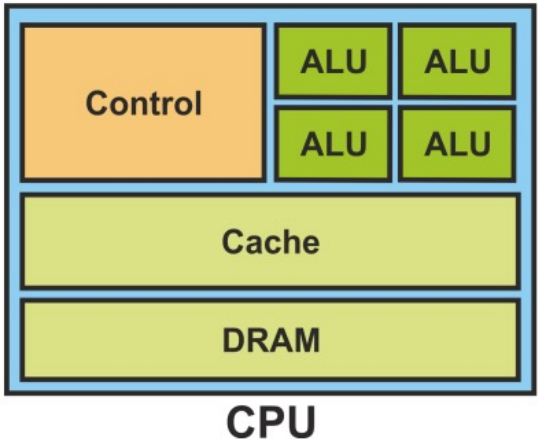
*However, constant dielectric environment gives significant artifacts for heterogeneous dielectric environment*

*Ex) proteins solvated in water, membrane solvated in water*

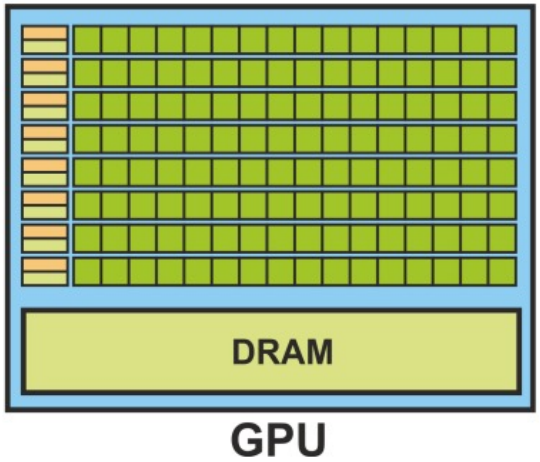
***Real system : nonuniform counter charge distribution, generates electrostatic gradients***
***PME w/ non-neutral system : No electrostatic gradient***

# 4. GPU parallelization of Nonbonded Energy Computation

## CPU vs GPU




- Central Processing Unit
- Serial cores
- Low latency
- Good for serial processing
- Can do handful of operation at once




- Graphic Processing Unit
- Many cores
- High throughput
- Good for parallel processing
- Can do thousands of operation at once

*Paz, A., & Plaza, A.,* A new morphological anomaly detection algorithm for hyperspectral images and its GPU implementation. In Satellite Data Compression, Communications, and Processing VII(Vol. 8157)., *SPIE*, **2011**

# 4. GPU parallelization of Nonbonded Energy Computation

## GPU Acceleration

*It is based on the principle of parallel computing*

*Scalability is the important factor to make the parallel computing better*

**Table 1. Summary of Comparisons between Different Electrostatic Force Calculation Algorithms (EFC Alg.)[a]**

| EFC Alg. | complexity | accuracy | scalability | boundary conditions |
|---|---|---|---|---|
| DCS | $O(NQ)$[60] | Highest[47] | Low[47] | NPC, PC |
| MSM | $O(N + Q)$[33] | Medium[33] | High[33] | PC, SPC, NPC |
| $P^3$ M | $O(N \log N)$[67] | High[32] | Medium[69,70] | PC |
| PME | $O(N \log N)$[32] | Medium[60,61] | High[62,63] | PC |
| FMM | $O(N)$[79,80] | Medium[77,78,82] | High[81] | NP, PC |

[a]NPC stands for nonperiodic conditions, PC for periodic conditions, and SPC for semiperiodic conditions.

*George, A., Mondal, S., Purnaprajna, M., & Athri, P.,* Review of Electrostatic Force Calculation Methods and Their Acceleration in Molecular Dynamics Packages Using Graphics Processors. *ACS omega,* ***2022****, 7(*37*),* 32877-32896.

## *Takeaways*

- For computing energy in molecular simulation box, PBC *is applied to prevent excess surface effect in the system. Sometimes artificial environment from PBC induce artifacts of simulation*

- Several algorithms to compute high-cost nonbonded energy were developed(Cutoff, PME, PPPM, FMM, Reaction Field).

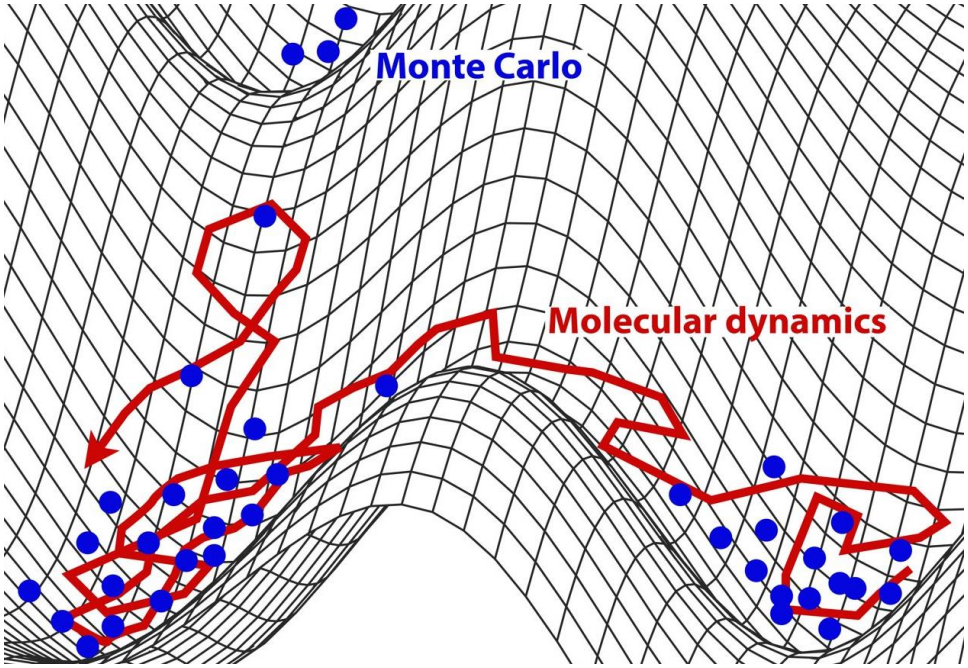
- Nowadays, PME is the most used algorithm for nonbonded energy calculation due to its performance. However, PME has several limitations and artifacts.

- GPU can accelerate nonbonded computation. To utilize GPU for it, scalability of computing algorithms matters.

*Thank you for your kind attention / Q&A*

# *Monte Carlo Basics*

*Week 2 : How to Sample Ensembles?*




2024 Winter Molecular Simulation Seminar

January 26th, 2024

Presenter: Seonghyeon Kang

Advisor: Prof. Chang Yun Son

# *Week 2 : How to Sample Ensembles?*

## *Three Brief Approaches to Sample Ensemble Average*

***1. Sampling in Molecular Dynamics Simulation: Thermostat***

***2. Sampling in Monte Carlo : Monte Carlo Basics***

***3. Enhanced Methods with Monte Carlo : Enhanced Monte Carlo***

## *Contents*


***1. What is Monte Carlo Simulation and How does it work?***

***2. Monte Carlo Sampling Methods in Various Ensembles***

***3. Grand Canonical Monte Carlo : Principles and Applications***

***4. Monte Carlo Simulation in Gibbs Ensemble***

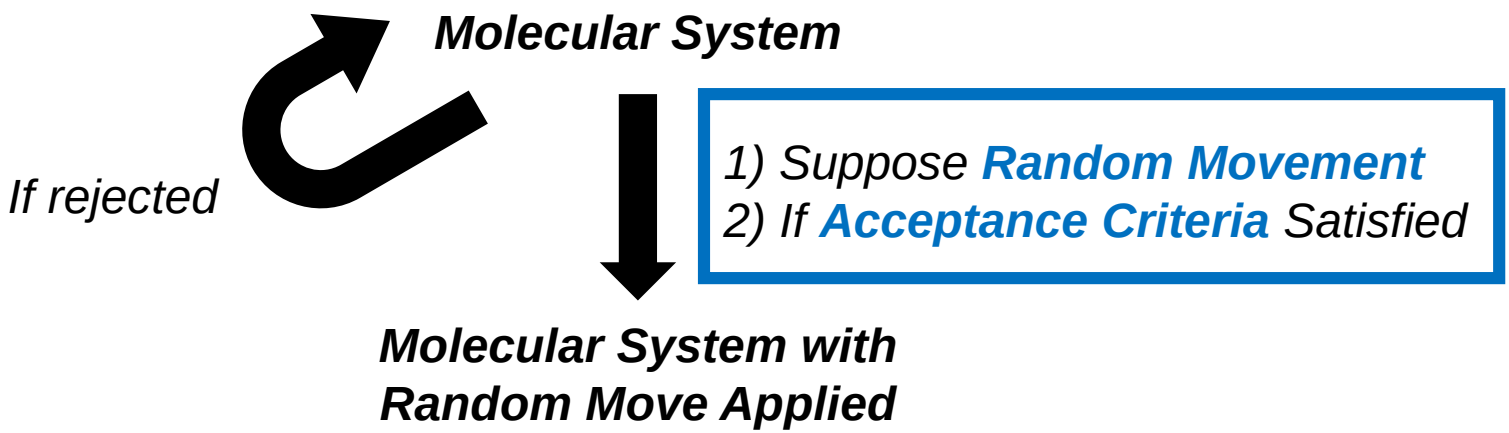
# 1. *What is Monte Carlo Simulation* and *How does it work*?

## *What is Monte Carlo Simulation?*

***Monte Carlo (MC) Methods*** *: Statistical methodologies that* ***derive certain statistical quantities for given system by utilizing random number***

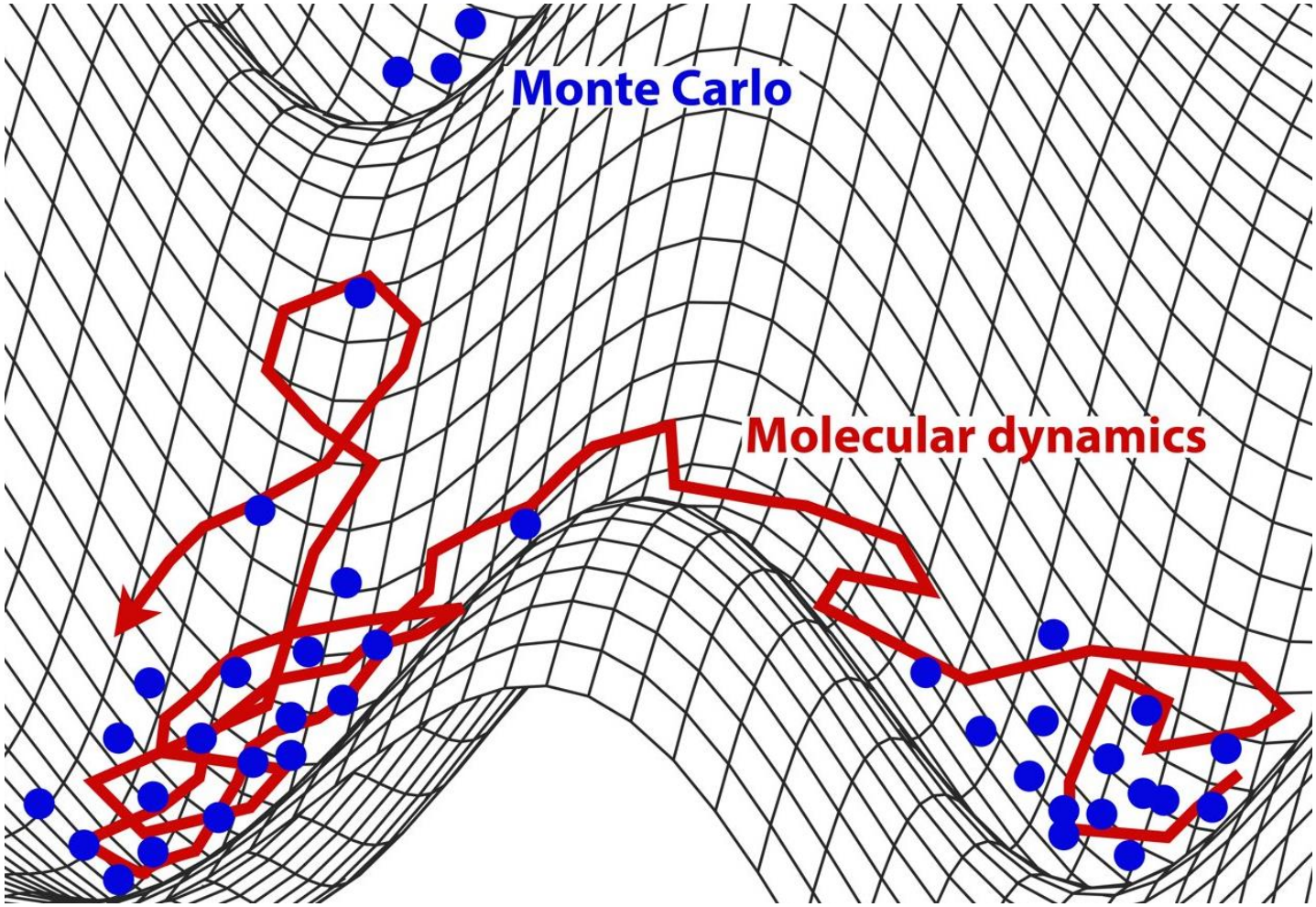
***Monte Carlo Simulation*** *: Molecular Simulation methodologies that use* ***the idea of Monte Carlo Method***

## *How Monte Carlo Simulation Works? – Metropolis Algorithm*



## *1. What is Monte Carlo Simulation and How does it work?*

### *Monte Carlo Simulation vs Molecular Dynamics Simulations*



# *1. What is Monte Carlo Simulation and How does it work?*

## *Why We Use Monte Carlo Simulation?*

### *Unique Perspective of Monte Carlo Simulation*

#### *1. Suggesting Random Movement in Chemical System*

*- It can also introduce artificial movement that MD cannot*

*- Particle insertion, Phase Transition, Hole hopping, …*

*- Mainly treated in this talk*

#### *2. Introducing Acceptance Criteria to Derive Specific Statistics*

*- It can derive the accepted states to certain probability distribution*

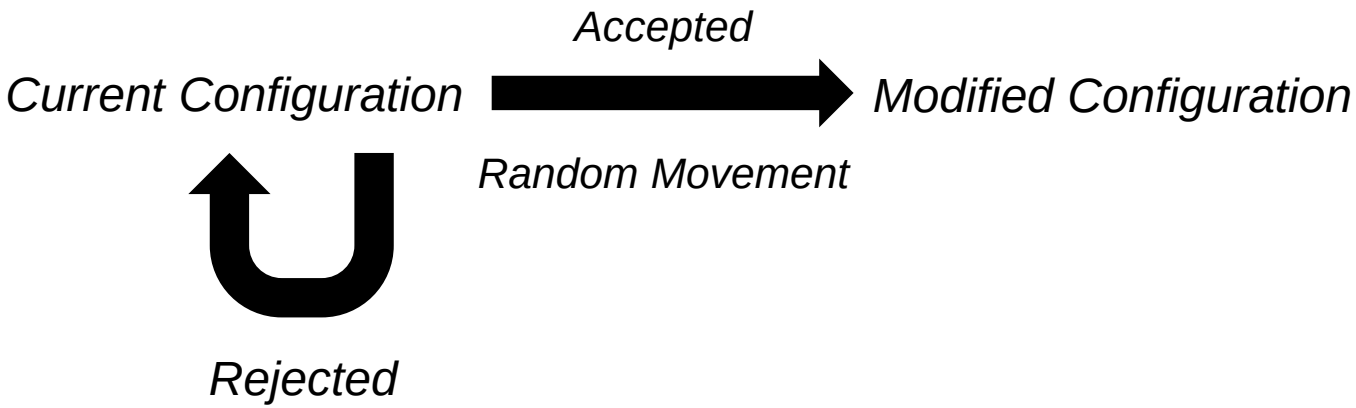
*- Ensemble expansion, Enhancing performance of MC or MD, …*

*- Mainly treated in the next talk(Enhanced Monte Carlo)*

## *2. Monte Carlo Sampling Methods* ***in Various Ensembles***

***Types of Ensembles : NVE, NVT, NPT, …***

***NVE Ensemble(Microcanonical Ensemble)***




***<Acceptance Criteria>***
***If $\Delta E < 0$ : Accept the random movement***
***If $\Delta E > 0$ : Reject the random movement***

***"Exactly the same as energy minimization process!"***

*Frenkel, D., & Smit, B.*, Understanding molecular simulation: from algorithms to applications. *Elsevier*. **2023**

# 2. Monte Carlo Sampling Methods in Various Ensembles

**Types of Ensembles : NVE, NVT, NPT, …**

## NVT Ensemble(Canonical Ensemble)

*Current Configuration* $\xrightarrow{\text{Accepted}}$ *Modified Configuration*

*Random Movement*

*Rejected*

**<Acceptance Criteria>**

**If $\Delta E < 0$ : Accept the random movement**

**If $\Delta E > 0$ : Accept with the probability $e^{-\Delta E/k_b T}$**

**Else : Reject**

$$acc(i \rightarrow j) \Rightarrow min\left(1, exp\left(-\frac{\Delta E}{k_b T}\right)\right)$$

$$\underbrace{\frac{P(state\ 2)}{P(state\ 1)} = \frac{e^{-\Delta E/k_b T}}{1}}_{\text{In Statistical Mechanics (Equilibrium)}} = \underbrace{\frac{acc(1 \rightarrow 2)}{acc(2 \rightarrow 1)}}_{\text{In Monte Carlo. (Equilibrium)}} \Rightarrow P(state\ 1)acc(1 \rightarrow 2) = P(state\ 2)acc(2 \rightarrow 1)$$

**Detailed Balance (Equilibrium Statistics)**

*Frenkel, D., & Smit, B.*, Understanding molecular simulation: from algorithms to applications. *Elsevier*. **2023**

# 2. Monte Carlo Sampling Methods in Various Ensembles

**Types of Ensembles : NVE, NVT, NPT, …**

## NPT Ensemble(Isobaric-Isothermal Ensemble)

*Accepted*

*Current Configuration* → *Modified Configuration*

*Random Movement*

*Rejected*

- ***Detailed Balance* Also Applied**
- ***Energy difference due to volume change* should be applied**

**<Acceptance Criteria>**

**If $\Delta W = \Delta E + P\Delta V - Nk_bT\Delta(\log(V)) < 0$ : Accept the random movement**

**If $\Delta W = \Delta E + P\Delta V - Nk_bT\Delta(\log(V)) > 0$ : Accept with the probability $e^{-\Delta W/k_bT}$**

**Else : Reject**

$$acc(i \rightarrow j) \Rightarrow min\left(1, exp\left(-\frac{\Delta E + P\Delta V - Nk_bT\Delta(\log(V))}{k_bT}\right)\right)$$

*Frenkel, D., & Smit, B.*, Understanding molecular simulation: from algorithms to applications. *Elsevier*. **2023**

# 2. Monte Carlo Sampling Methods in Various Ensembles

***Types of Ensembles : NVE, NVT, NPT, …***

## *NPT Ensemble(Isobaric-Isothermal Ensemble)*

**Ex) MonteCarloBarostat in OpenMM(2 MD steps + 1 MC Step)**

$$r(t+\mathrm{d}t) = r(t) + v(t)\ \delta t + \tfrac{1}{2}a(t)\ \delta t^2,$$
$$v\left(t+\tfrac{1}{2}\delta t\right) = v(t) + \tfrac{1}{2}a(t)\ \delta t,$$
$$ma(t+\delta t) = -\nabla E[r(t+\delta t)],$$
$$v(t+\delta t) = v\left(t+\tfrac{1}{2}\delta t\right) + a(t+\delta t)\ \delta t,$$

***Two MD steps***

$$V' = V + R[S(\delta V)] \qquad r'(t+\delta t) = r(t+\delta t)\left[\frac{V'^{1/3}}{V^{1/3}}\right]$$

***One MC step(Trial Move)***

$$P(\Delta V) = \begin{cases} \exp\left(-\dfrac{\Delta W}{kT_0}\right), & \Delta W > 0, \\ 1, & \Delta W \le 0. \end{cases}$$

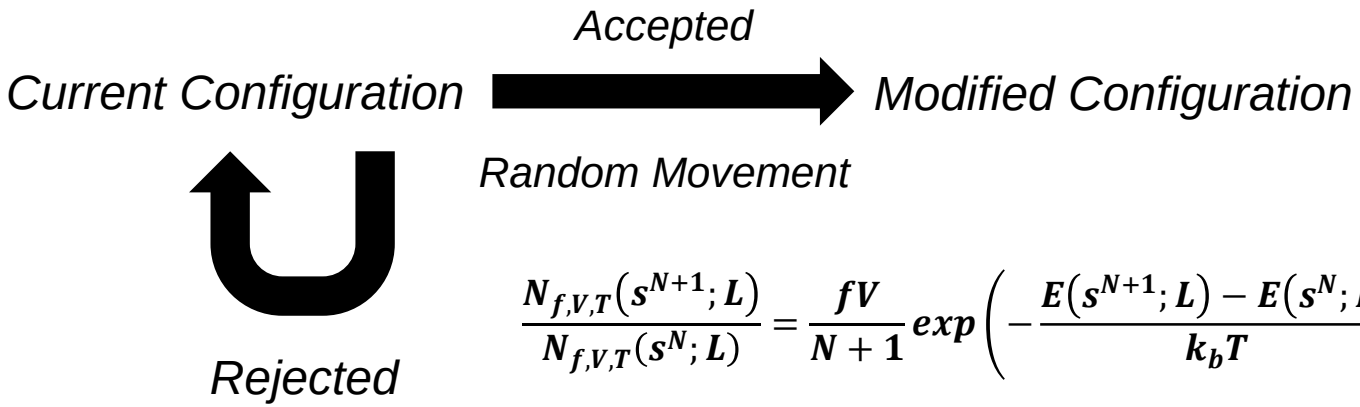
$$\Delta W = (E' - E) + P_0(V' - V) - NkT_0\ \ln\frac{V'}{V}$$

***One MC step (Acceptance Criteria)***

*Åqvist, J., Wennerström, P., Nervall, M., Bjelic, S., & Brandsdal, B. O.,* Molecular dynamics simulations of water and biomolecules with a Monte Carlo constant pressure algorithm. *Chem. Phys. Lett.*, **2004**, *384*(4-6), 288-294.

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Principle : muVT Ensemble(Grand Canonical Ensemble)




$$\frac{N_{f,V,T}(s^{N+1};L)}{N_{f,V,T}(s^{N};L)} = \frac{fV}{N+1} exp\left(-\frac{E(s^{N+1};L) - E(s^{N};L)}{k_b T}\right)$$

### <Acceptance Criteria>

**1) Displacement of Particles :** $acc(i \rightarrow j) \Rightarrow min\left(1, exp\left(-\frac{\Delta E}{k_b T}\right)\right)$

**2) Insertion of Particles :** $acc(N \rightarrow N+1) \Rightarrow min\left(1, \frac{fV}{N+1} exp\left(-\frac{\Delta E(N+1) - \Delta E(N)}{k_b T}\right)\right)$

**3) Removal of Particles :** $acc(N \rightarrow N-1) \Rightarrow min\left(1, \frac{N}{fV} exp\left(-\frac{\Delta E(N+1) - \Delta E(N)}{k_b T}\right)\right)$

### How to calculate the energy per molecule(Chemical Potential)?

*Frenkel, D., & Smit, B.*, Understanding molecular simulation: from algorithms to applications. *Elsevier*. **2023**

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Principle : muVT Ensemble(Widom's Particle Insertion Method)

$$Q_N = \frac{1}{N!}\int_V \cdots \int_V \; exp\left(-\frac{E_N}{k_bT}\right) dr_1\, dr_2 \cdots dr_N$$

$$= \int \; \frac{1}{(N-1)!}\left(\int_V \cdots \int_V \; exp\left(-\frac{E_{N-1}}{k_bT}\right) dr_1\, dr_2 \cdots dr_{N-1}\right) exp\left(-\frac{\psi}{k_bT}\right) dr_N$$

$$= Q_{N-1}\frac{V}{N}\left\langle exp\left(-\frac{\psi}{k_bT}\right)\right\rangle$$

*$\psi$ is the interaction energy of an inserted particle with all other particles in the system*

$$\mu_{insert} = \frac{\Delta F}{\Delta N} = \frac{F(N) - F(N-1)}{1} = -k_bTlog\left(\frac{Q_N}{Q_{N-1}}\right) = -k_bTlog\left(\frac{1}{\rho}\left\langle exp\left(-\frac{\psi}{k_bT}\right)\right\rangle\right)$$

## Widom's Particle Insertion Method

$$\mu_{insert} = -k_bTlog\left(\frac{1}{\rho}\left\langle exp\left(-\frac{\psi}{k_bT}\right)\right\rangle\right)$$

*Widom. B.*, Some Topics in the Theory of Fluids, *J. Chem. Phys.*, **1963**, 39(11), 2808-2812
*Widom. B.*, Potential-distribution theory and the statistical mechanics of fluids. *J. Phys. Chem.*, **1982**, *86*(6), 869-872.

# 3. Grand Canonical Monte Carlo : Principles and Applications

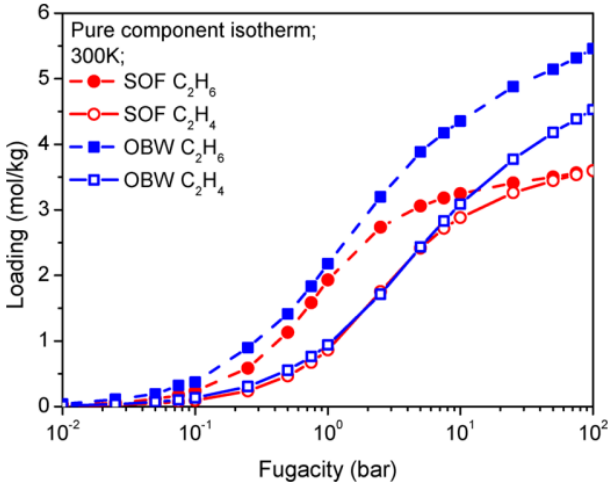
## Application: Gas Adsorption

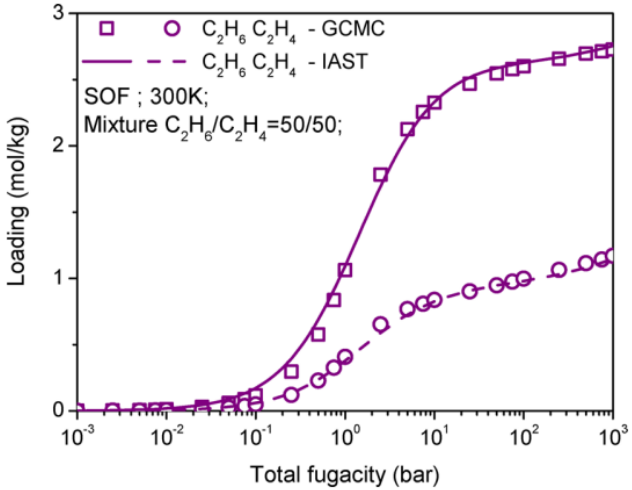
***GCMC allowed the molecular simulation with exchange of matter***

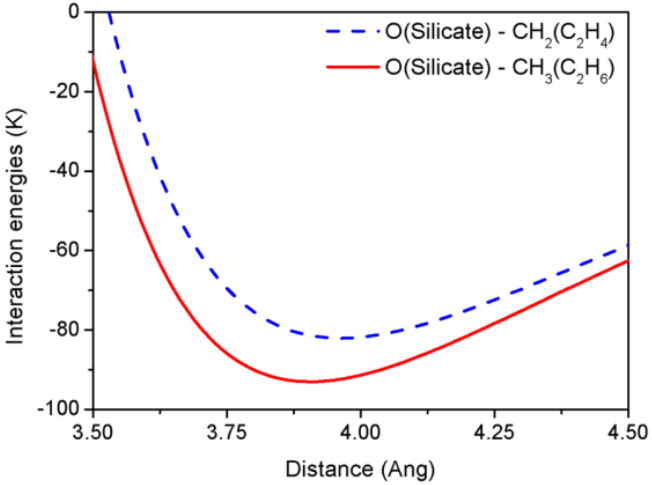
***It enabled to applying GCMC to gas adsorption in porous material***

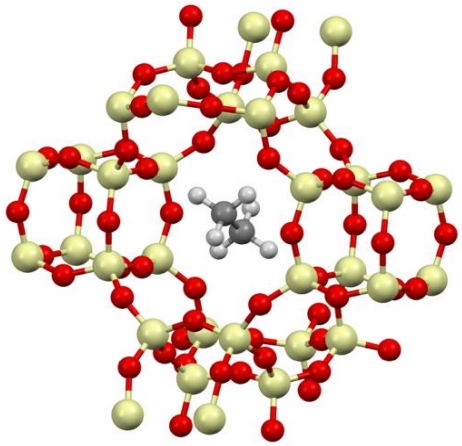
***Computed Isotherm + Interaction by using GCMC***

*Kim, J., Lin, L. C., Martin, R. L., Swisher, J. A., Haranczyk, M., & Smit, B.,* Large-scale computational screening of zeolites for ethane/ethene separation. *Langmuir,* **2012**, *28(32), 11914-11919.*



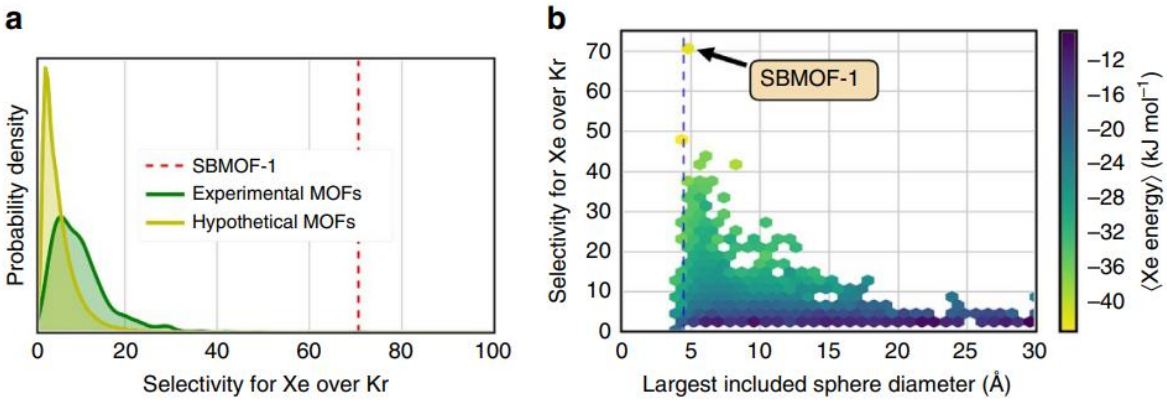



# 3. Grand Canonical Monte Carlo : Principles and Applications

## Application: Gas Adsorption

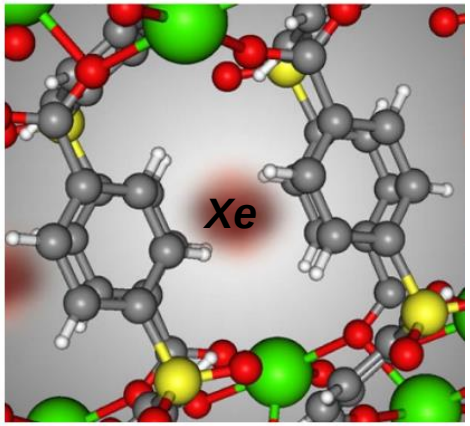
### Grand Canonical Monte Carlo + Widom Particle Insertion Method

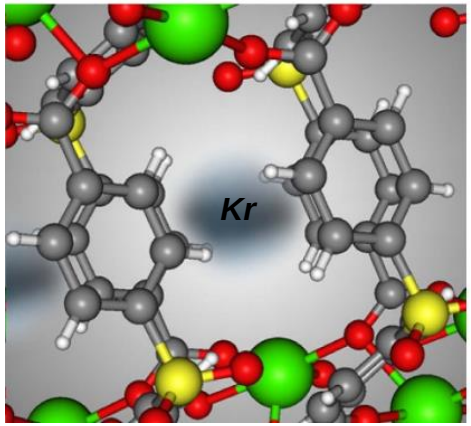



### MOF Screening

$$N_i(p_i) = K_{H,i} p_i (i = Xe, Kr)$$

$$K_H = \left\langle e^{-\frac{U}{RT}} \right\rangle / RT$$



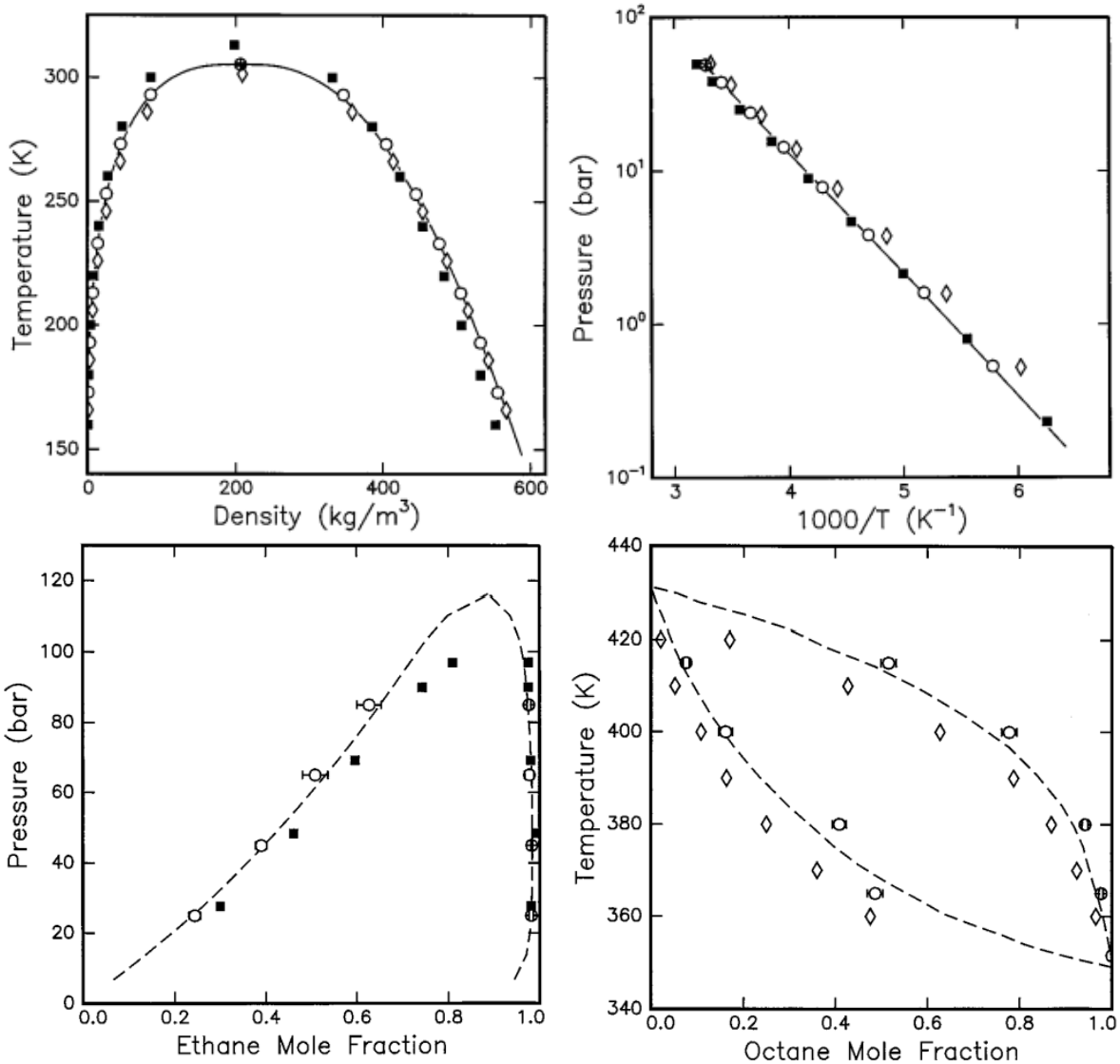




### Spatial probability densities sampled from GCMC

*Banerjee, D., Simon, C. M., Plonka, A. M., Motkuri, R. K., Liu, J., Chen, X., ... & Thallapally, P. K.*, Metal–organic framework with optimally selective xenon adsorption and separation. *Nat. Comm.*, **2016**, *7*(1), ncomms11831.

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Application: Vapor-Liquid Equilibrum(Histogram Reweighting)

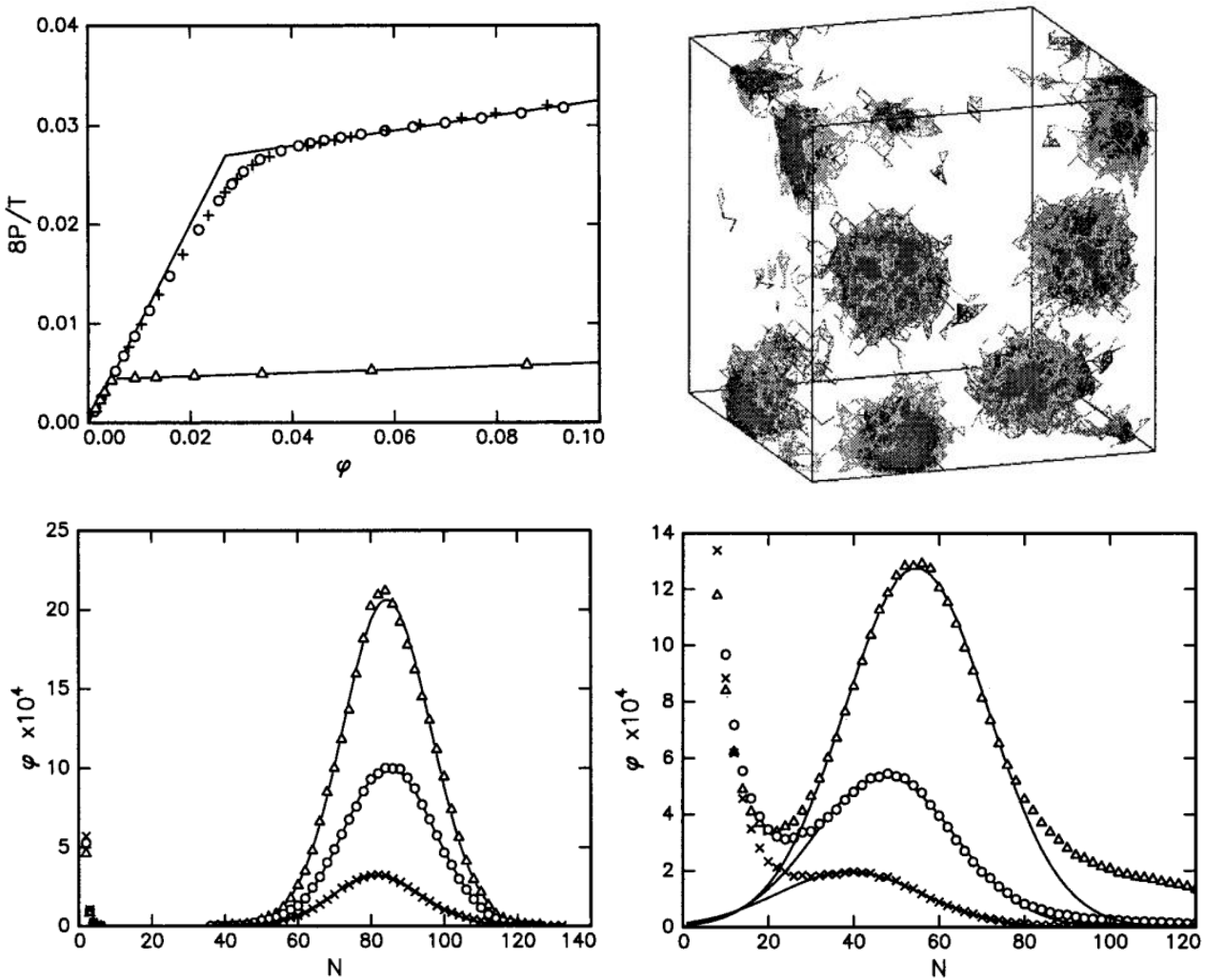



***With histogram reweighting method(Next seminar), Vapor-Liquid Equilibrium also can be modelled***

*Errington, J. R., & Panagiotopoulos, A. Z.,* A new intermolecular potential model for the n-alkane homologous series. *J. Phys. Chem. B*, **1999**, *103*(30), 6314-6322.

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Application: Critical Micelle Concentration (Histogram Reweighting)




*Floriano, M. A., Caponetti, E., & Panagiotopoulos, A. Z*. Micellization in model surfactant systems. *Langmuir*, **1999**, *15*(9), 3143-3151.

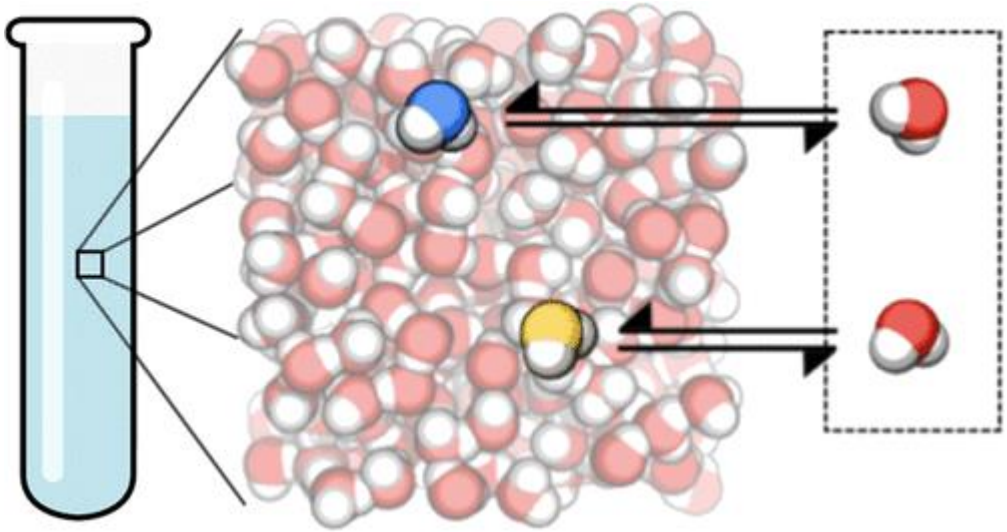
# 3. Grand Canonical Monte Carlo : Principles and Applications

## Technical Issues : Molecule Insertion Problem

***In perspective of sampling, acquiring sufficient insertion/deletion move in the phase space is important***

***However, acceptance ratio for molecule insertion/deletion is significantly low!(Especially high density & low temperature system)***

*Soroush Barhaghi, M., Torabi, K., Nejahi, Y., Schwiebert, L., & Potoff, J. J.* Molecular exchange Monte Carlo: A generalized method for identity exchanges in grand canonical Monte Carlo simulations. *J. Chem. Phys.*, **2018**, *149*(7).



*Ross, G. A., Rustenburg, A. S., Grinaway, P. B., Fass, J., & Chodera, J. D.,* Biomolecular simulations under realistic macroscopic salt conditions. *J. Phys. Chem. B*, **2018**. *122*(21), 5466-5486.

## 3. Grand Canonical Monte Carlo : Principles and Applications

### Solution : Molecular Exchange Monte Carlo

**2. Solutions**

**Configuration-Bias Monte Carlo, Continuous Fractional Component Monte Carlo, Expanded Ensembles, …**

**- (Controlling acceptance criteria, Next talk : Enhanced Monte Carlo)**

**Identity exchange move(swapping molecule A and B, Controlling Move)**

**- Steric issue can be solved. Many simulations (Grand canonical, Semi-grand canonical, Gibbs ensemble, …) use this idea**

**- However, this method is difficult to generalize in various system**

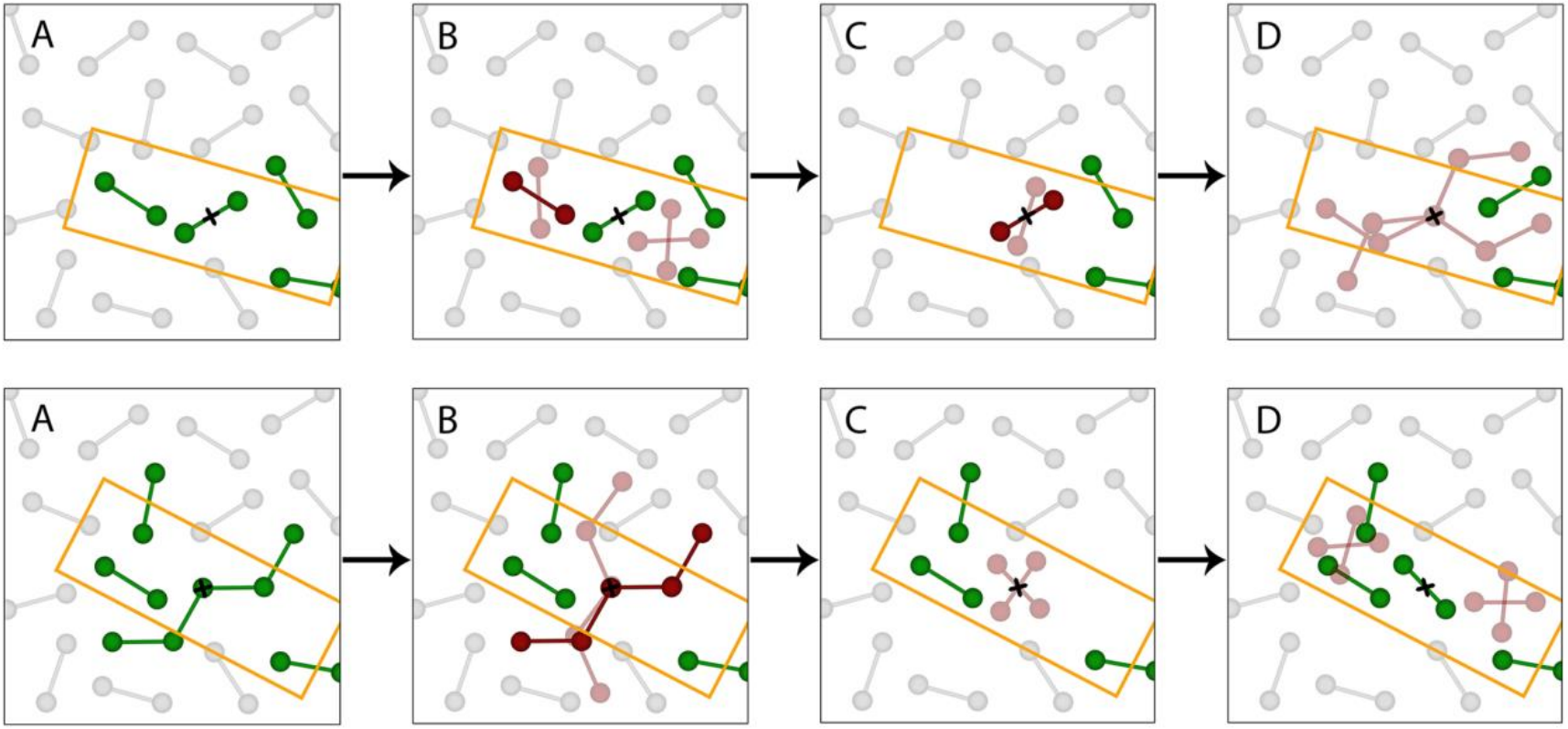
**- Solution : Generalized identity exchange move for simulations in the grand canonical ensemble(Molecular Exchange Monte Carlo)**

*Soroush Barhaghi, M., Torabi, K., Nejahi, Y., Schwiebert, L., & Potoff, J. J.* Molecular exchange Monte Carlo: A generalized method for identity exchanges in grand canonical Monte Carlo simulations. *J. Chem. Phys.*, **2018**, *149*(7).

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Solution : Molecular Exchange Monte Carlo

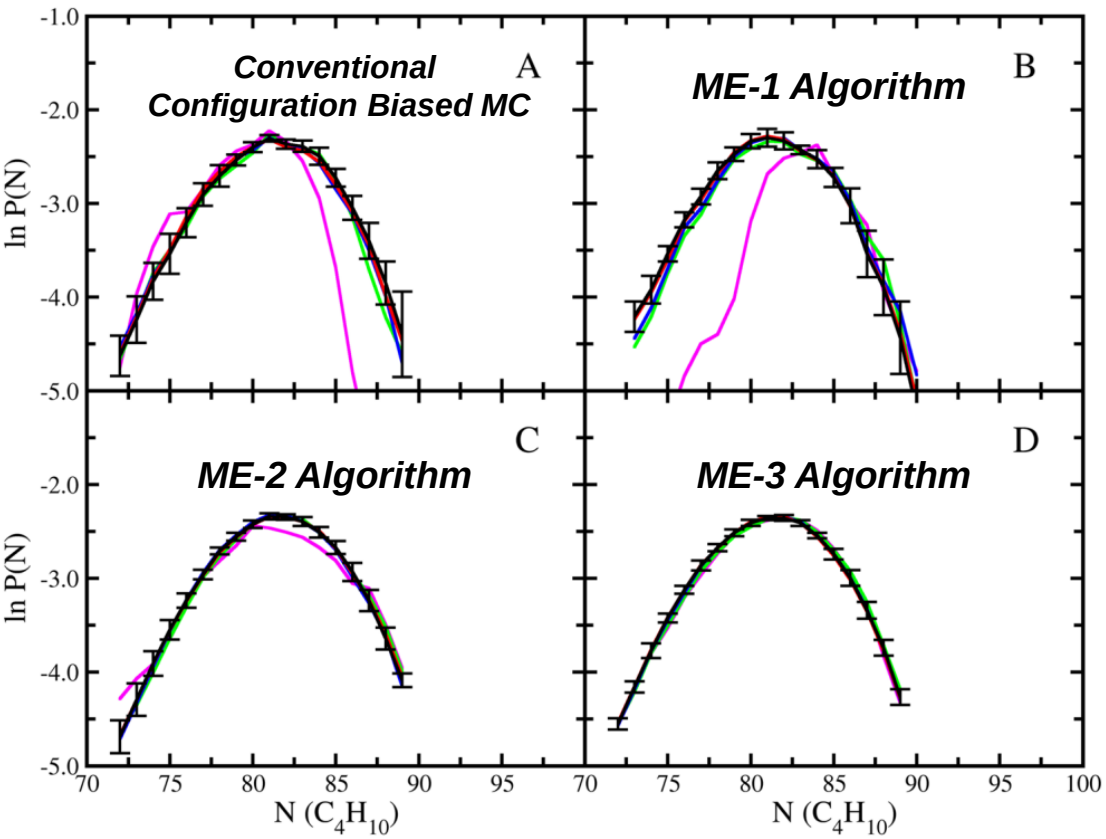
### ME-1, ME-2, ME-3 Algorithm

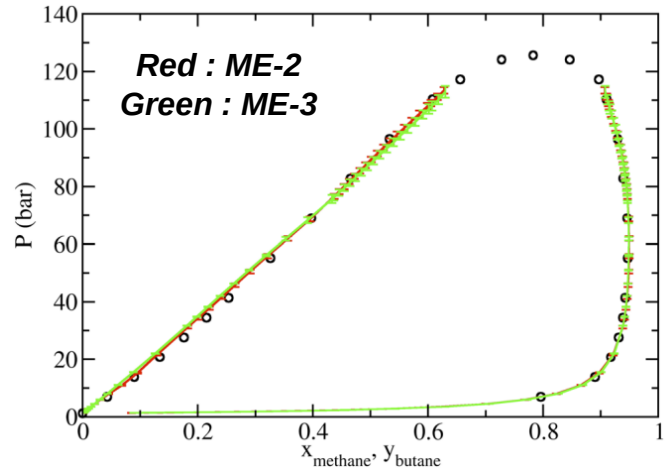


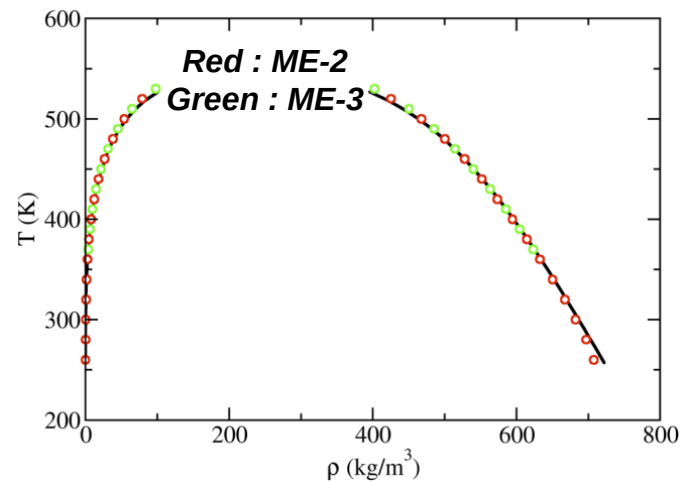

*Soroush Barhaghi, M., Torabi, K., Nejahi, Y., Schwiebert, L., & Potoff, J. J.* Molecular exchange Monte Carlo: A generalized method for identity exchanges in grand canonical Monte Carlo simulations. *J. Chem. Phys.*, **2018**, *149*(7).

# 3. Grand Canonical Monte Carlo : Principles and Applications

## Solution : Molecular Exchange Monte Carlo



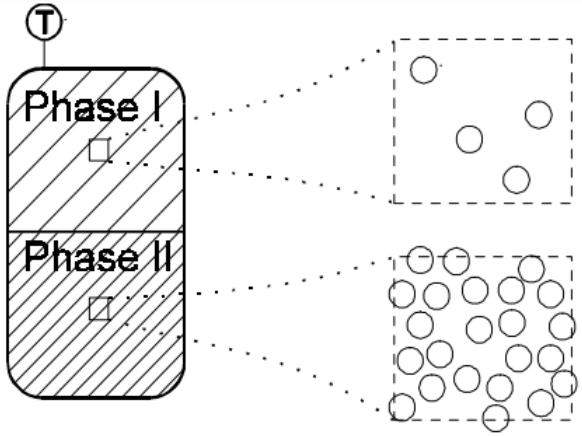




*Soroush Barhaghi, M., Torabi, K., Nejahi, Y., Schwiebert, L., & Potoff, J. J.* Molecular exchange Monte Carlo: A generalized method for identity exchanges in grand canonical Monte Carlo simulations. *J. Chem. Phys.*, **2018**, *149*(7).

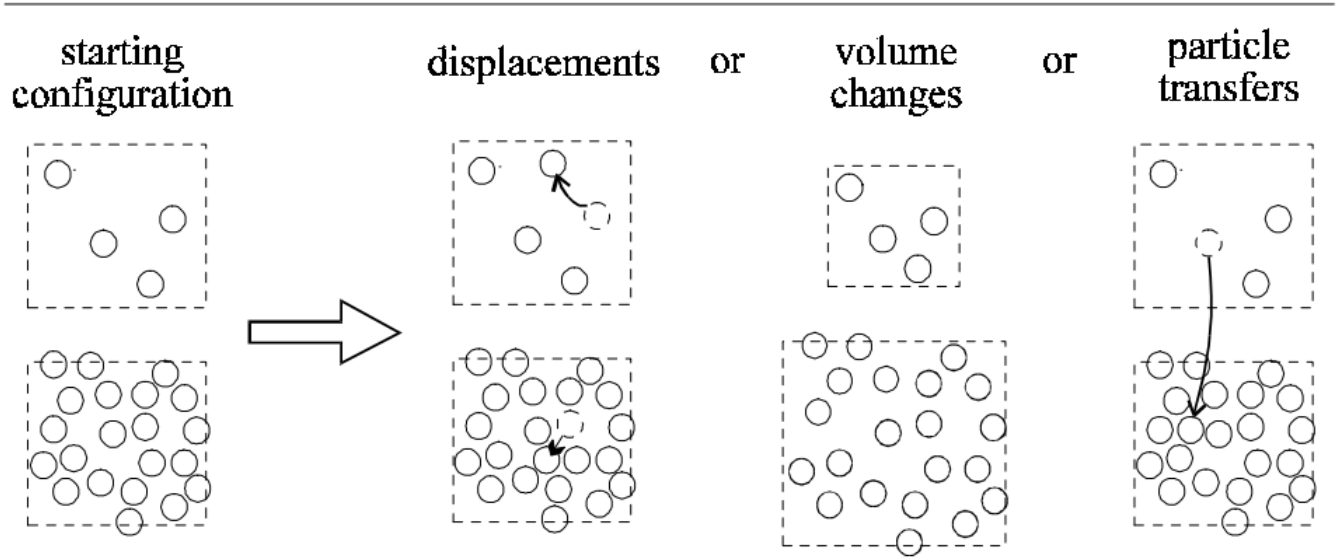
# 4. Monte Carlo Simulation in Gibbs Ensemble

## Principle : Phase Coexistence without Boundary

$$\mu_I = \mu_{II}$$
$$V_I + V_{II} = const.$$
$$T_1 = T_2$$



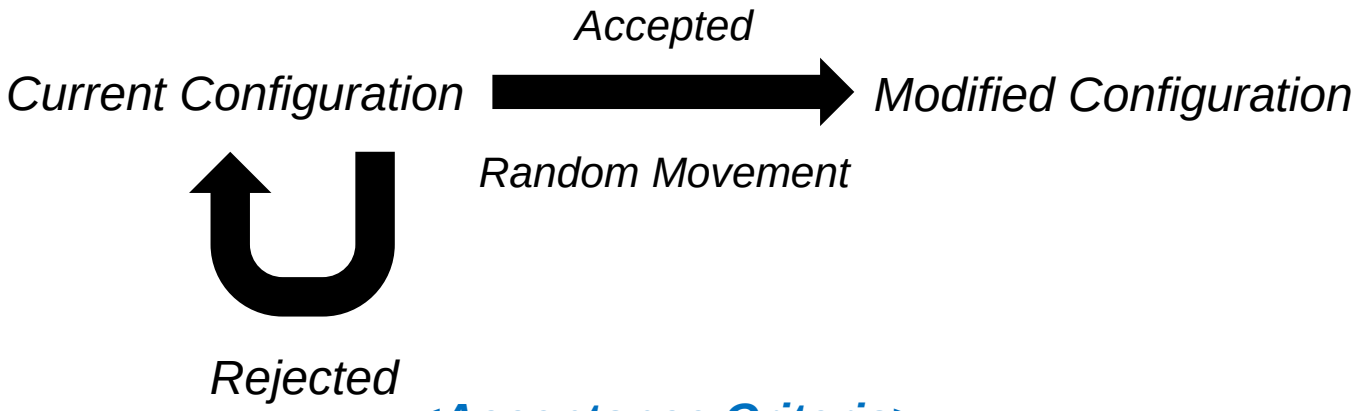

**But it is not μPT Ensemble!**




*Baus, M., Rull, L. F., & Ryckaert, J. P. (Eds.)*., Observation, prediction and simulation of phase transitions in complex fluids (Vol. 460)., Springer Science & Business Media, **2012**

# 4. Monte Carlo Simulation in Gibbs Ensemble

## Principle : Acceptance Criteria




### <Acceptance Criteria>

**1) Displacement :** $P_{move} = min\left(1, exp\left(-\frac{\Delta E}{k_b T}\right)\right)$

**2) Volume Change :**

$$P_{volume} = \min\left[1, \exp\left(-\frac{\Delta U_I + \Delta U_2}{k_b T} + N_I \log\left(\frac{V_I + \Delta V}{V_I}\right) + N_{II}\, log\left(\frac{V_{II} - \Delta V}{V_{II}}\right)\right)\right]$$

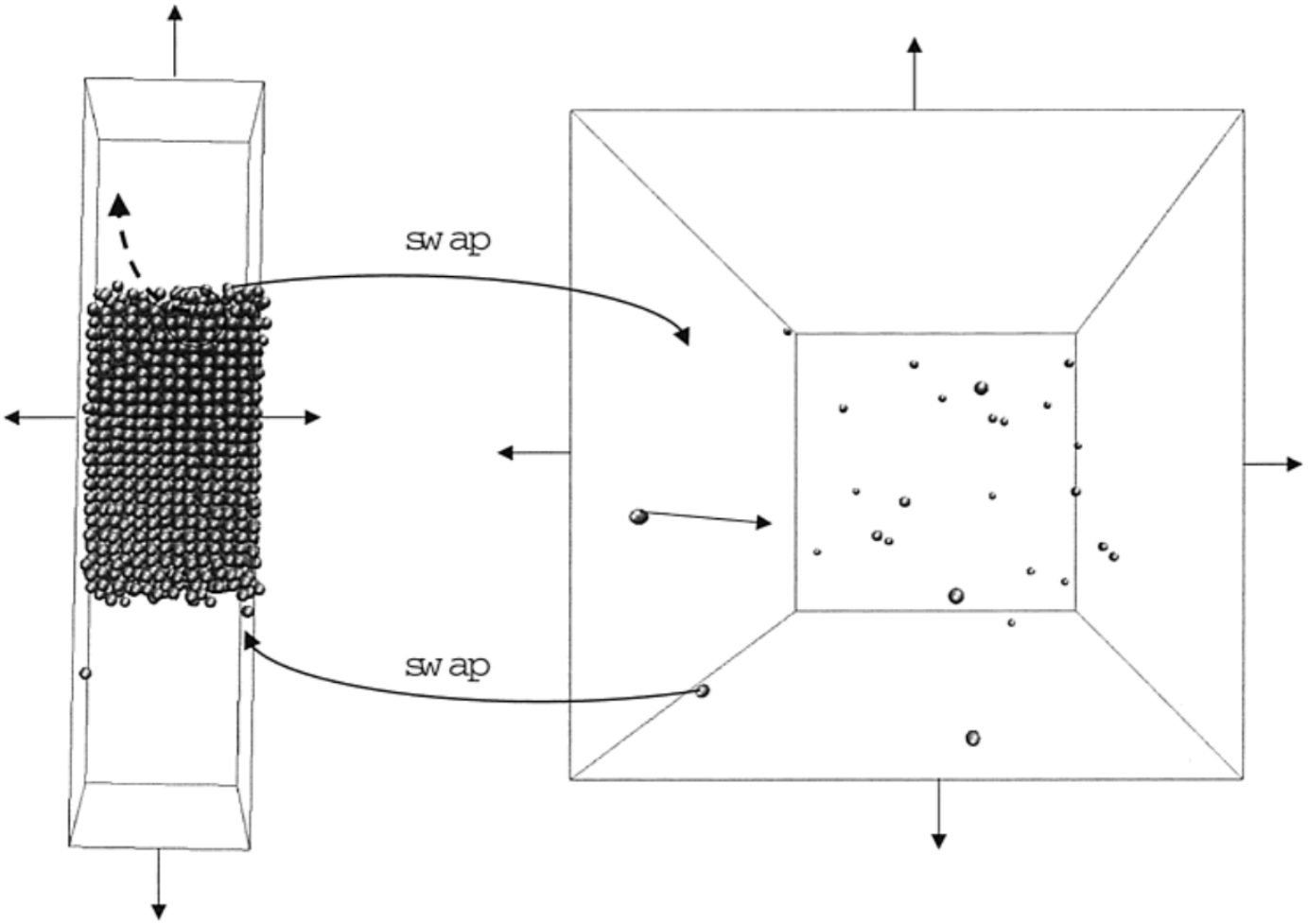
$\Delta V = \xi \delta v_{max} min(V_I, V_{II})$ $(0 < \xi < 1, random, \delta v_{max} : maximum\ volume\ fraction)$

**3) Particle Transfer(I to II):** $P_{transfer} = \min\left[1, \frac{N_{II} V_I}{(N_{II}+1) V_I} \exp\left(-\frac{\Delta U_I + \Delta U_2}{k_b T}\right)\right]$

*Baus, M., Rull, L. F., & Ryckaert, J. P. (Eds.)*., Observation, prediction and simulation of phase transitions in complex fluids (Vol. 460)., *Springer Science & Business Media*, **2012**

# 4. Monte Carlo Simulation in Gibbs Ensemble

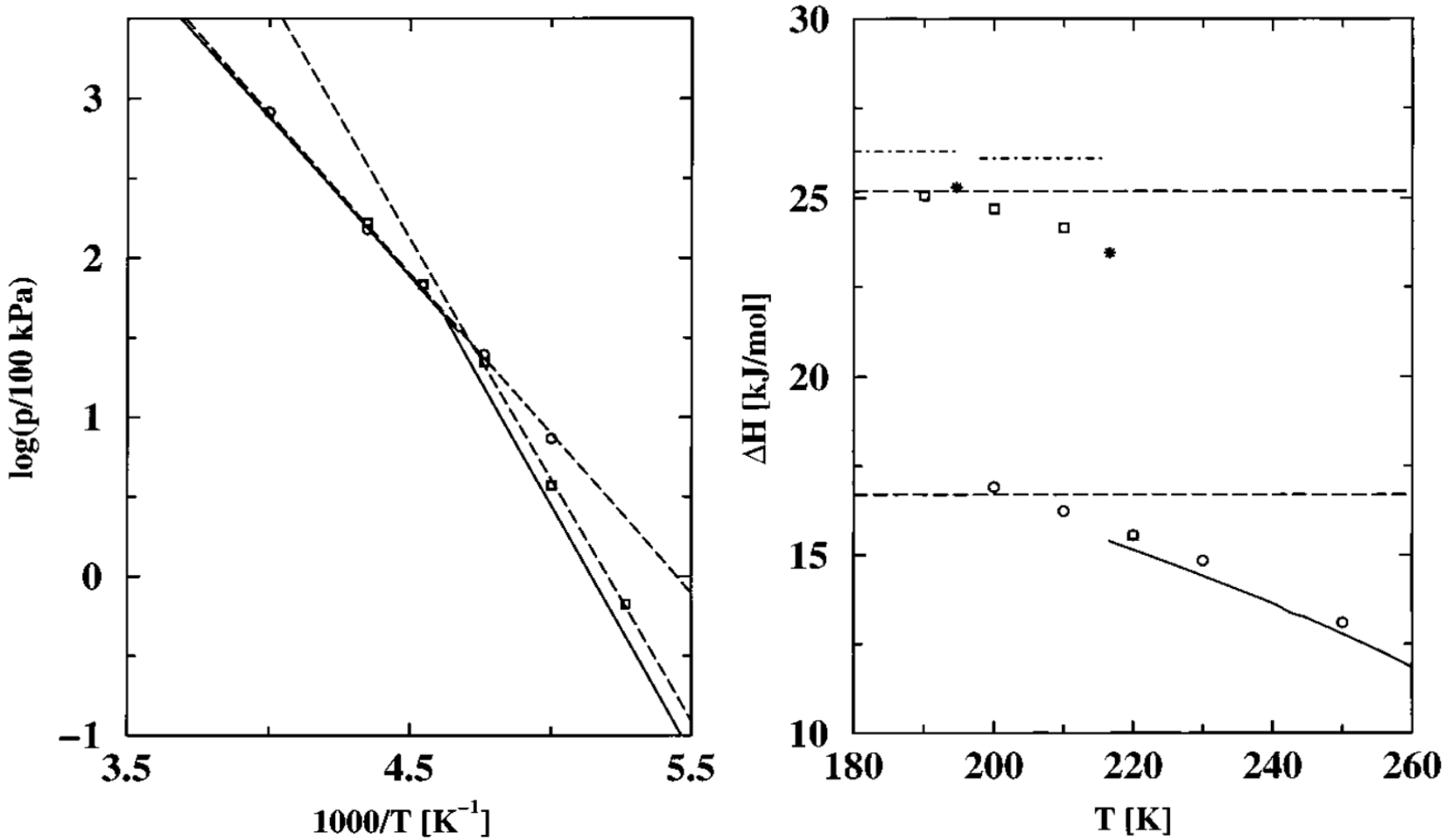
## Application : Solid – Vapor Equilibrium




*Chen, B., Siepmann, J. I., & Klein, M. L.,* Direct Gibbs ensemble Monte Carlo simulations for solid− vapor phase equilibria: applications to Lennard− Jonesium and carbon dioxide. *J. Phys. Chem. B*, **2001**, *105*(40), 9840-9848.

# 4. Monte Carlo Simulation in Gibbs Ensemble

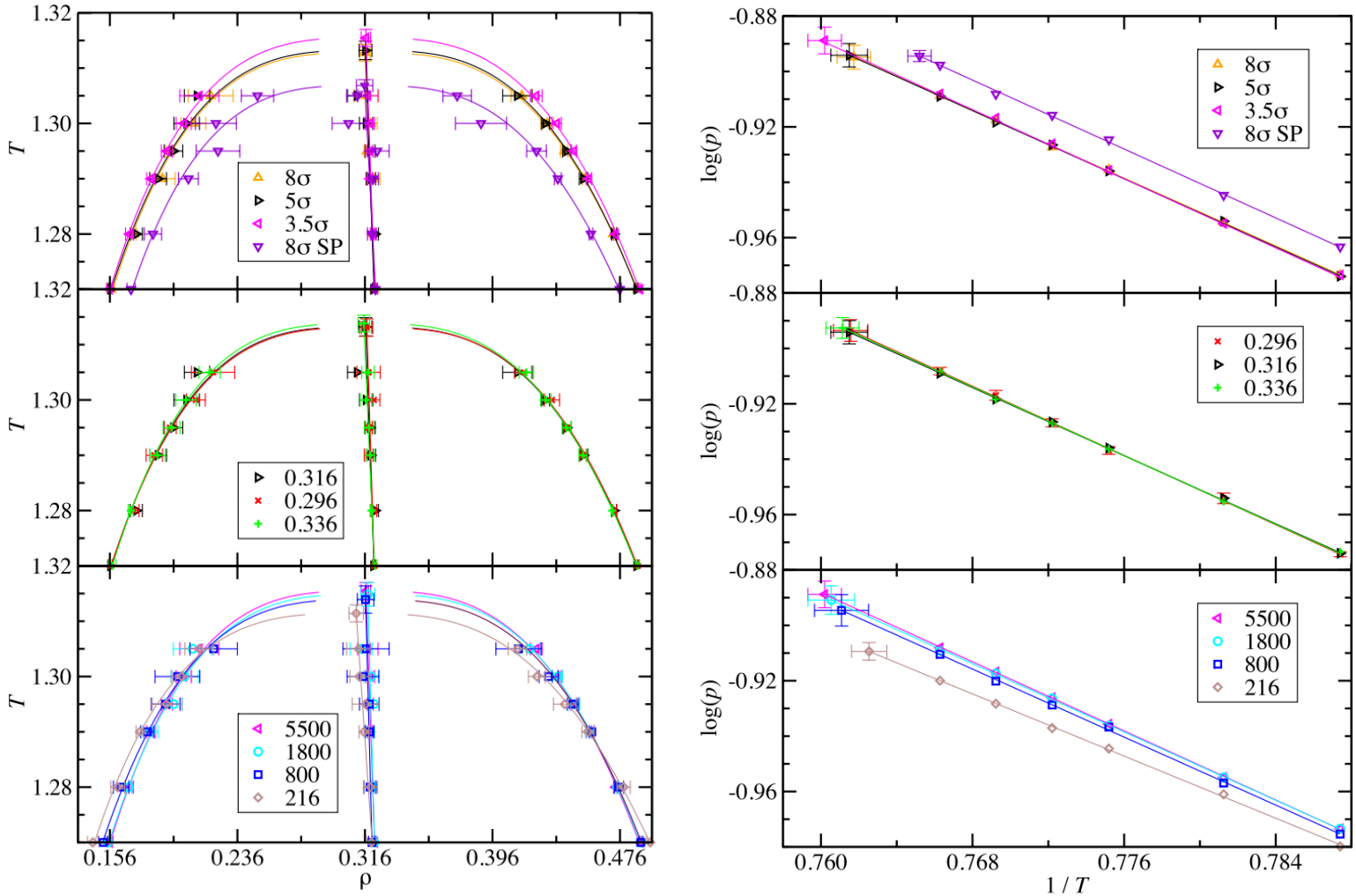
## Application : Solid – Vapor Equilibrium




*Chen, B., Siepmann, J. I., & Klein, M. L.,* Direct Gibbs ensemble Monte Carlo simulations for solid− vapor phase equilibria: applications to Lennard− Jonesium and carbon dioxide. *J. Phys. Chem. B*, **2001**, *105*(40), 9840-9848.

# 4. Monte Carlo Simulation in Gibbs Ensemble

## Application : Liquid – Vapor Equilibrium, Critical Point




*Dinpajooh, M., Bai, P., Allan, D. A., & Siepmann, J. I.*, Accurate and precise determination of critical properties from Gibbs ensemble Monte Carlo simulations. *J. Chem. Phys.*, **2015**, *143*(11).

## *Takeaways*

■ *Monte Carlo method is statistical methodology that derive certain statistical quantities for given system by utilizing random number. It contains two main processes : Random Movement and Acceptance Criteria*

■ *By using specific acceptance criteria, we can sample the system with NVE, NVT, and NPT ensemble environment*

■ *Grand Canonical Monte Carlo allowed the molecular simulation with exchange of matter. It contains several principles related to particle insertion and deletion(Acceptance criteria, Widom's particle insertion method, …)*

■ *Gibbs ensemble assume two-phase system as separated systems with particle swapping. It enabled to modelling phase transition phenomena*

*Thank you for your kind attention / Q&A*